

Periods and L -functions of meromorphic modular forms via finite contours

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Abstract

In this note, we synthesize and generalize previous work, which (1) extracted periods for weak cusp forms in a manifestly finite manner [1], (2) identified a non-holomorphic regularization prescription to define finite analogs of L-functions associated to weakly holomorphic modular forms [2], and (3) identified a contour prescription to define an L-function for meromorphic modular forms with poles inside the fundamental domain [3]. Our construction, derived from the cohomological period extraction of weak cusp forms in [1], shows that these three lines of work not only naturally align, but derive from the *same* manifestly finite linear functional acting on the space of $\mathrm{SL}_2(\mathbb{Z})$ modular forms.

We call this linear functional the L -integral, and denote it by $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$, where $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ are paths from $S^{-1}\tau_0, T^{-1}\tau_0$, respectively, to τ_0 . When $f \in M_k$, $L^*(f, s) = \Gamma(s) \sum_{n \neq 0} \frac{c_f(n)}{(2\pi n)^s}$: it is *the* classical L-function. Our L-integral is finite and τ_0 -independent (within each admissible contour homotopy class) for any meromorphic modular form f .

We use this L-integral to (1) extract published numerical values of the periods of the weak cusp form $\widehat{\Delta} := \frac{1}{q} + O(q^2) \in S_{12}^1$ in [1], (2) recover the regularized expressions for the divergent Mellin transform of f in [2], and (3) recover the L-functions of [3] for meromorphic forms with interior poles that are regular at the cusps. When f has poles inside the fundamental domain, $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is unique and constant for paths within the same homotopy class; related $L^*(f, s)$ -integrals along paths in different homotopy classes differ by wall-crossing residues. Finally, all major results were tested to arbitrary (50 digit) precision in our python scripts [4].

1 Introduction and main results

In this paper, we introduce a general framework that defines L-functions for generic modular forms on the full group $\mathrm{SL}_2(\mathbb{Z})$, and extracts their periods for the explicit case when the holomorphic cusp forms are one-dimensional, i.e. when $\dim(S_k) = 1$. Classically, L-functions and periods are both well-defined and well-studied for holomorphic modular forms. However, there is a growing body of work exploring periods [1] and L-functions [2] for weakly holomorphic modular forms. Similarly, there are natural physical motivations to consider L-functions for properly meromorphic modular forms, which have poles *within* the fundamental domain [3].

In these works, and specifically in [2, 3], and throughout physics, divergences due to singularities in quantities of interest necessitate the introduction of regulators in order to define finite quantities. Regulators, though, often break symmetries, and care is needed to check that the regulator has not injected undue violence to the system. Of relevance here is that [2] introduced a non-holomorphic regulator to deal with the singularities at either end of the classical Mellin contour, $\int_0^\infty$, sufficient to regulate divergences at the cusps, but which violates holomorphy in the deformed contour-integral, ruining path-independence. Simultaneously, [3] concerned itself with L-functions for modular forms which diverge within the interior of the fundamental domain. Here, holomorphic dependence on τ is paramount, which puts this treatment at odds with the regulator in [2].

In this note, we introduce manifestly finite contour integral functionals to define L-functions for generic meromorphic modular forms that are of sufficient generality to handle any combination of poles at cusps and poles within the fundamental domain. Our construction is motivated by, and makes precise contact with, work which extracts period-coefficients from weak cusp forms [1]. Our integrals are manifestly finite throughout, and reproduce all results in the literature, unifying the quasi-period results and the L-function extensions within one concrete, natural framework.

What we construct. Let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, let f be a meromorphic modular form of even weight $k \in \mathbb{Z}$ on Γ , and let $\tau_0 \in \mathbb{H}$ be a basepoint with the property that the pair of paths

$$\begin{cases} S\text{-segment:} & \gamma_{\tau_0}^S : [0, 1] \rightarrow \mathbb{H}, & \gamma_{\tau_0}^S(0) = -1/\tau_0, \gamma_{\tau_0}^S(1) = \tau_0, \\ T\text{-segment:} & \gamma_{\tau_0}^T : [0, 1] \rightarrow \mathbb{H}, & \gamma_{\tau_0}^T(0) = \tau_0 - 1, \gamma_{\tau_0}^T(1) = \tau_0, \end{cases} \quad (1)$$

in $\mathbb{H}$ avoids the Γ -orbit of the poles of f in $\mathbb{H}$. Using these contours, we define

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := i^{-s} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right], \quad (2)$$

where the T -kernel is built from the following explicit two Hurwitz-zeta functions:

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1). \quad (3)$$

When the contour avoids poles in f , both contour integrals contributing to $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ in (2) are finite for every basepoint $\tau_0 \in \mathbb{H}$ admissible for f and every $s \in \mathbb{C}$; no regulator, asymptotic-subtraction argument, or analytic-continuation step is required. Further, as we will show, $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is invariant under both τ_0 -variations and $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ -variations, as long as neither $\gamma_{\tau_0}^S$ nor $\gamma_{\tau_0}^T$ cross any pole location within the fundamental domain.

We will show that $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is the classical L-function for $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$. Further, because of its τ_0 -independence and finiteness for $f \in M_k^!(\mathrm{SL}_2(\mathbb{Z}))$, and because we reproduce the analytic formulae in [2], we unambiguously show that their candidate L-function is the canonical L-function one would associate with any weak modular form, $f \in M_k^!$.

Three theorems. The functional $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ obeys three properties that together identify it as the canonical finite-contour L-function of f :

- (i) (*Theorem 1.1*) When $\dim S_k = 1$, the integer- s specialisations of (2) reduce to a polynomial-kernel form that produces the periods $\omega^\pm(f)$ and quasi-periods $\eta^\pm(f)$ as an explicit linear functional in the Fourier coefficients of f . For $f = \Delta$ at $k = 12$ this recovers the Brown values of $\omega^\pm(\Delta)$; for $f = \hat{\Delta}$ this gives the quasi-periods $\eta^\pm(\hat{\Delta})$ from a finite-endpoint contour pair, with no divergence at the cusp.
- (ii) (*Theorem 1.2*) For $f \in M_k^!$, $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is τ_0 -independent on $\mathbb{H}$ and produces the completed L-function $L^*(f, s)$ at every $s \in \mathbb{C}$ as a single closed-form incomplete-gamma sum, agreeing term by term in the Fourier expansion with [2]. Our proof in fact exploits the simplicity at the elliptic fixed-point basepoint $\tau_0 = i$, where the S -segment collapses to a point and (2) reduces to a single integral on the compact contour $[i-1, i]$.
- (iii) (*Theorem 1.3*) For general meromorphic f , $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ depends on $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ only through their homotopy class in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, with explicit wall-crossing residue jumps when the homotopy class changes by crossing a pole. Restricted to forms regular at the cusp, this reproduces the deformed-Mellin L-function $L_f^*(s)_{\mathrm{int}}$ (Corollary 1.4; see [3]).

The complete statements are below; their proofs occupy §3–§5.

Theorem 1.1 (Periods and quasi-periods as an explicit linear functional, $\dim S_k \leq 1$). *Let $k \in 2\mathbb{Z}$ with $\dim S_k \leq 1$, and let $f \in S_k^!$. For the substantive cases $k \in \{12, 16, 18, 20, 22, 26\}$ where $\dim S_k = 1$, the period and quasi-period coefficients $\omega^\pm(f)$ are given by the explicit two-segment linear functional*

$$\omega^\pm(f) = \frac{(2\pi i)^{k-1} (-1)^{\ell-1} \binom{k-2}{\ell-1}}{c_S^{\ell-1}} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, \ell) d\tau \right] \quad (4)$$

for every $f \in S_k^!$, every $\tau_0 \in \mathbb{H}$, and every interior $\ell \in \{2, \dots, k-2\}$, with parity $\pm$ determined by $\ell - 1 \bmod 2$. Here $c_S^{\ell-1} := [p_f^\pm]_{X^{n-(\ell-1)}Y^{\ell-1}} \in \mathbb{Q}$ is the Brown cuspidal-basis coefficient at the matched monomial ([1, 13]), $\tilde{k}_T(\tau, s)$ is the Hurwitz-zeta T -kernel of (3), and its integer- $s = \ell$ specialisation collapses via $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$ ($m \geq 0$) to the $\mathbb{Q}$ -rational polynomial

$$\tilde{k}_T(\tau, \ell) = -\frac{B_\ell(\tau+1)}{\ell} + (-1)^{\ell-1} \frac{B_{k-\ell}(\tau+1)}{k-\ell} \in \mathbb{Q}[\tau] \quad (5)$$

of degree at most $k-2$. The right side of (4) is independent of τ_0 and of the choice of interior ℓ within its parity sector.

Inside the two-segment bracket of (4) sit only the bare Mellin monomial $\tau^{\ell-1}$ on $\gamma_{\tau_0}^S$ and the bare Hurwitz polynomial $\tilde{k}_T(\tau, \ell)$ on $\gamma_{\tau_0}^T$; the factors $(2\pi i)^{k-1}$, $(-1)^{\ell-1}$, $\binom{k-2}{\ell-1}$ and the Brown basis coefficient $c_S^{\ell-1}$ are absorbed into the prefactor.

Higher-dimensional S_k . For $k \geq 24$ with $\dim S_k \geq 2$, the Brown cuspidal-basis coefficient $c_S^{\ell-1}$ in (4) becomes matrix-valued: each $\ell \in \{2, \dots, k-2\}$ gives a $\dim S_k$ -component linear functional on $\{c_f(n)\}$ rather than a scalar. The structural argument behind Theorem 1.1 — coboundary inversion on V_{k-2} via δ_T^{-1} in closed Bernoulli-operator form — generalises without conceptual change, but the explicit kernel tables grow with $\dim S_k$, and the ℓ - and τ_0 -independence statement of Corollary 3.8 becomes a matrix identity. Working out the explicit matrix-valued kernels for $k = 24, 28, 30, \dots$ and connecting them to the existing Eichler–Shimura matrix-period basis literature is the natural generalisation of Theorem 1.1, and we leave it to a follow-up. Theorems 1.2 and 1.3 are unaffected and apply at every even k .

Theorem 1.2 (L -function for weakly holomorphic forms). *Let $k \in 2\mathbb{Z}$ and $f \in M_k^!$. With $\tilde{k}_T(\tau, s)$ the Hurwitz-zeta kernel of (3), for every $\tau_0 \in \mathbb{H}$ and every $s \in \mathbb{C}$,*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := i^{-s} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right] \quad (6)$$

is finite and independent of both $\tau_0 \in \mathbb{H}$ and of the admissible contour choices $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$. We write $L^*(f, s)$ for the common value.

Specialising to the elliptic fixed point $\tau_0 = i$ of the modular involution $S: \tau \mapsto -1/\tau$, the S -segment $[-1/i, i]$ collapses to the single point i , the S -integral in (6) vanishes, and L^* reduces to a single integral on the compact contour $[i-1, i] \subset \mathbb{H}$, with closed-form sum

$$L^*(f, s) = i^{-s} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right]. \quad (7)$$

The right side of (7) is the completed L -function $L^*(f, s)$ of f . For holomorphic modular forms it reduces to the classical Hecke–Weil Mellin transform. For weak modular forms, it reduces to the results of [2], proving that their non-holomorphic regulator prescription gives the unique, canonical L -function for every weak modular form.

Classical Mellin recovery and canonical uniqueness. For $f \in S_k$ a holomorphic cusp form, taking the cusp-limit $\tau_0 \rightarrow i\infty$ in (6) collapses the S -segment onto the classical Mellin contour $[0, i\infty]$ while the T -segment vanishes exponentially (since $f(\tau) = O(e^{-2\pi \operatorname{Im} \tau})$ on the segment as $\operatorname{Im} \tau \rightarrow \infty$):

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = \int_0^\infty t^{s-1} f(it) dt, \quad (8)$$

the classical completed L -integral of f . For $f \in M_k$ with $c_f(0) \neq 0$ the same limit holds after a standard constant-term subtraction (see [3, Lemma 2.1] for the explicit polynomial correction).

For $f \in M_k^!$ with cusp poles the cusp-limit is genuinely pathological: $f(\tau) \rightarrow \infty$ as $\operatorname{Im} \tau \rightarrow \infty$, so the S -segment integral diverges along the imaginary axis, and the T -segment integral $\int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau$ acquires a matching divergence of the opposite sign as $\operatorname{Im} \tau_0 \rightarrow \infty$. The two divergences cancel at every finite τ_0 , so $L^*(f, s)$ is finite and well defined throughout $\mathbb{H}$, but no naive Mellin contour limit exists. The construction of [2] recovers $L^*(f, s)$ on $M_k^!$ by a non-holomorphic regulator at the cusps; Theorem 1.2 shows that this regulator and the finite-contour functional (6) compute the same object, term by term in the Fourier expansion. In particular, $L^*(f, s)$ is independent of any choice of regularisation: it is the value of the finite-contour functional at any admissible $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$, and the BFK construction is one closed-form evaluation of it (at $\tau_0 = i$).

Theorem 1.3 (*L -function for meromorphic modular forms*). *Let $k \in 2\mathbb{Z}$ and let f be a meromorphic modular form of weight k on Γ . Write $\mathbb{H}_f := \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ for the upper half-plane with the Γ -orbit of the interior poles of f removed, and call the pair $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ admissible if $\tau_0 \in \mathbb{H}_f$ and both contours lie in $\mathbb{H}_f$, with $\gamma_{\tau_0}^S$ running from $S^{-1}\tau_0 = -1/\tau_0$ to τ_0 and $\gamma_{\tau_0}^T$ from $T^{-1}\tau_0 = \tau_0 - 1$ to τ_0 . Define $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ by (2) with the Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ of (3). Then:*

- (a) *Both integrals in (2) are finite for every admissible $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ and every $s \in \mathbb{C}$.*
- (b) *$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ depends on the admissible pair only through its homotopy class in $\mathbb{H}_f$: any continuous one-parameter family of admissible pairs — including variations of the anchoring basepoint τ_0 — leaves the value unchanged.*
- (c) *If two admissible configurations $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ and $(\gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T)$ lie in homotopy classes related by a single crossing of an interior pole $\tau_p \in \mathbb{H}$ of f , then*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) - L^*(f, s; \gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T) = \mathcal{R}_{\tau_p}(f, s), \quad (9)$$

where for a simple pole $\operatorname{Res}(f, \tau_p)$,

$$\mathcal{R}_{\tau_p}(f, s) = -2\pi i i^{-s} \operatorname{Res}(f, \tau_p) \left[\sigma_{\tau_p}^S \tau_p^{s-1} + \sigma_{\tau_p}^T \tilde{k}_T(\tau_p, s) \right], \quad (10)$$

with $\sigma_{\tau_p}^S, \sigma_{\tau_p}^T \in \{-1, 0, +1\}$ the winding-number contributions of $\gamma_{\tau_0}^S$ and $\gamma_{\tau_0}^T$ across τ_p (the generic case has one nonzero and the other zero, according to which contour crossed). Higher-order poles contribute via the corresponding derivatives in τ_p of (10); the closed form is given in §5.

Corollary 1.4 (L^* on regular-at-cusps meromorphic forms). *Let $f \in F_k$ be a meromorphic modular form of weight k that is regular at the cusp. Let $L_f^*(s)_{\text{int}, \gamma}$ denote the deformed-Mellin L -function of f along a contour γ from 0 to $i\infty$ lying in $\mathbb{H}_f$ (see [3, §2]). Then for every admissible $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ whose homotopy class in $\mathbb{H}_f$ is compatible with γ (in the sense that the contour $\gamma_{\tau_0}^S$ together with the cusp-limit piece $\lim_{\tau_0 \rightarrow i\infty} \gamma_{\tau_0}^S \rightarrow [0, i\infty]$ can be deformed to γ in $\mathbb{H}_f$),*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = L_f^*(s)_{\text{int}, \gamma} \quad \text{for every } s \in \mathbb{C}. \quad (11)$$

The structural picture. The contour pair (1) has two natural one-parameter specialisations of the basepoint, each exploiting a fixed point of the modular curve $Y(1)$: the cusp specialisation $\tau_0 \rightarrow i\infty$ sends the S -segment to the classical Mellin contour $[0, i\infty]$ and the T -segment to 0 exponentially; the elliptic specialisation $\tau_0 = i$ kills the S -segment identically and leaves L^* as a single integral on $[i-1, i]$. The cusp specialisation, applied at finite $\tau_0 = i\Lambda$, gives the Λ -truncated Mellin contour of $L_f^*(s)_{\text{int}}$ (see [3]) as the S -segment alone, with the T -segment serving as the missing ingredient that produces the cocycle and renders the $\Lambda \rightarrow \infty$ limit unnecessary. The Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ in (3) is the closed form of the T -segment kernel: it is the unique V_n -analog, valid at every $s \in \mathbb{C}$, of the polynomial coboundary inversion of [7] (which itself recovers as the integer- s specialisation, by Theorem 1.1).

Outline. §2 introduces the period-defining cocycle and its moment expansion in the polynomial coefficient system V_{k-2} . §3 identifies the bare Mellin and Hurwitz integer- s specialisations as the period-extracting kernels and proves Theorem 1.1. §4 derives the Hurwitz-zeta kernel $\tilde{k}_T$ via a coboundary equation, proves Theorem 1.2 via a Hurwitz-unfolding lemma, and contains an explicit worked example at $k = 12$ exhibiting the Bernoulli-polynomial collapse of $\tilde{k}_T$ at integer s together with numerical values of $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$. §5 proves Theorem 1.3 and Corollary 1.4. §6 collects conceptual framings (holonomy/Wilson-loop interpretation, on-shell-amplitudes analogy, the regularisation-as-symptom heuristic, the open question on the geometric content of the $\tau^{s-1} f d\tau$ Mellin kernel, and pointers to the sequel on systematic per-mode theory for interior-pole forms). Appendix A records the algebraic derivation of the Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ from contour-invariance of the two-segment functional: starting from the classical Mellin transform split at the contour endpoints $\{-1/\tau_0, \tau_0\}$ and demanding that the resulting incomplete-gamma tail be representable as a T -segment integral, one derives a shift-difference equation whose solution is the Hurwitz-zeta combination of (3). Appendix B is a brief Eichler–Shimura primer for readers approaching the subject from theoretical physics. This appendix is included for the working-version of this note and is intended to be removed before formal publication.

2 The period-defining cocycle and its moment expansion

The polynomial coefficient system. Let $V_n := \mathbb{C}[X, Y]_n$ be the space of complex homogeneous polynomials in two auxiliary variables X, Y of degree n , equipped with the right Γ -action

$$(P|_\gamma)(X, Y) = P(aX + bY, cX + dY) \quad \text{for } \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma.$$

For a meromorphic modular form f of weight k , write $n := k-2$ throughout and form the V_n -valued 1-form on $\mathbb{H}$

$$\omega_f(\tau) := f(\tau) (X - \tau Y)^n d\tau.$$

Modularity of f implies that ω_f is Γ -equivariant in the standard sense [13]: under $\tau \mapsto \gamma^{-1}\tau$, the weight- k transformation of f cancels the weight- $(-n)$ slash on $(X - \tau Y)^n$, so the V_n -valued 1-form ω_f is invariant under the simultaneous action of Γ on τ and on the auxiliary variables (X, Y) .

The construction of ω_f above uses only the weight- k modularity of f and makes no cuspidality assumption: ω_f is well-defined for every meromorphic modular form $f \in M_k^!$ (and indeed for meromorphic forms with poles in the fundamental domain, provided contours avoid the Γ -orbit of those poles), with the cuspidal hypothesis $f \in S_k^!$ entering only later via Lemma 3.1(c) as the solvability condition for the period-polynomial coboundary equation.

The cocycle. The two-segment cocycle construction used in the present note is the standard Eichler–Shimura cohomological setup [13], in which a weakly holomorphic modular form is integrated against the polynomial coefficient system $(X - \tau Y)^n$ along contours from $\gamma^{-1}\tau_0$ to τ_0 for the generators $\gamma \in \{S, T\}$ of Γ , with periods and quasi-periods extracted from the resulting 1-cocycle modulo coboundaries. The author was led to this presentation, and to its quasi-period reformulation, through personal correspondence with F. Brown [5]. Concretely: fix a basepoint $\tau_0 \in \mathbb{H}$ not on the Γ -orbit of any pole of f , and define

$$C_\gamma^f(\tau_0) := (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} \omega_f = (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} f(\tau) (X - \tau Y)^n d\tau \in V_n \otimes \mathbb{C}, \quad (12)$$

the contour being any path from $\gamma^{-1}\tau_0$ to τ_0 in $\mathbb{H}$ avoiding the Γ -orbit of the poles of f . The integral is independent of the choice of path within the same homotopy class in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$.

Lemma 2.1 (Cocycle property). *For all $g, h \in \Gamma$,*

$$C_{gh}^f(\tau_0) = C_g^f(\tau_0)|_h + C_h^f(\tau_0). \quad (13)$$

Equivalently, the map $\gamma \mapsto C_\gamma^f(\tau_0)$ is a 1-cocycle of Γ with coefficients in $V_n \otimes \mathbb{C}$.

Proof. Using $(gh)^{-1} = h^{-1}g^{-1}$ and path-additivity through the intermediate point $h^{-1}\tau_0$,

$$\int_{(gh)^{-1}\tau_0}^{\tau_0} = \int_{h^{-1}g^{-1}\tau_0}^{\tau_0} = \int_{h^{-1}g^{-1}\tau_0}^{h^{-1}\tau_0} + \int_{h^{-1}\tau_0}^{\tau_0}.$$

In the first integral on the right substitute $\tau = h^{-1}\sigma$, so σ runs from $g^{-1}\tau_0$ to τ_0 . The Γ -equivariance of ω_f converts the integrand to $\omega_f(\sigma)|_h$, yielding the first summand of (13). The second integral gives the second summand directly. $\square$

A cocycle on $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ is determined by its values on the two generators

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

subject to the relations $S^2 = (ST)^3 = -I$ (which act trivially on V_n for n even). We write

$$C_S^f(\tau_0) = (2\pi i)^{n+1} \int_{-1/\tau_0}^{\tau_0} \omega_f, \quad C_T^f(\tau_0) = (2\pi i)^{n+1} \int_{\tau_0-1}^{\tau_0} \omega_f, \quad (14)$$

since $S^{-1}\tau_0 = -1/\tau_0$ and $T^{-1}\tau_0 = \tau_0 - 1$. These are the two integrals over the contour pair (1).

Moment expansion. Expanding $(X - \tau Y)^n = \sum_{\ell=0}^n \binom{n}{\ell} (-\tau)^\ell X^{n-\ell} Y^\ell$ and pulling the polynomial out of the integral defines the moment integrals

$$I_S^\ell(f, \tau_0) := \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^\ell d\tau, \quad I_T^\ell(f, \tau_0) := \int_{\tau_0-1}^{\tau_0} f(\tau) \tau^\ell d\tau, \quad (15)$$

for $\ell = 0, 1, \dots, n$, in terms of which

$$C_S^f(\tau_0) = (2\pi i)^{n+1} \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell I_S^\ell(f, \tau_0) X^{n-\ell} Y^\ell,$$

and similarly for $C_T^f(\tau_0)$. Theorem 1.1 extracts $\omega^\pm(f)$ from particular linear combinations of these moments.

Basepoint dependence. Varying τ_0 changes C^f by a coboundary, provided f has no interior poles. Define an antiderivative $F(\tau)$ of ω_f on a simply connected open subset of $\mathbb{H}$ avoiding $\Gamma \cdot f^{-1}(\infty)$. Then for any two basepoints τ_0, τ'_0 in the same connected component, path-additivity gives

$$C_\gamma^f(\tau'_0) - C_\gamma^f(\tau_0) = (P|_\gamma - P) \quad \text{with} \quad P = (2\pi i)^{n+1} (F(\tau'_0) - F(\tau_0)) \in V_n \otimes \mathbb{C}. \quad (16)$$

This is the coboundary $\delta P(\gamma) := P|_\gamma - P$. When f has interior poles, dragging τ_0 across a pole picks up a Cauchy residue contribution beyond (16); this wall-crossing structure is the content of Theorem 1.3 in §5.

Cohomology and periods. The cohomology class $[C^f] \in H^1(\Gamma; V_n)$ is independent of τ_0 (on the component) and independent of the choice of δP representative. The classical Eichler–Shimura theorem [13] identifies the cuspidal subspace $H_{\text{cusp}}^1(\Gamma; V_n)$ with $S_k \oplus \overline{S_k}$, and the linear coordinates of $[C^f]$ in this cuspidal subspace are the periods and quasi-periods $\omega^\pm(f), \eta^\pm(f)$ [1].

3 Period extraction: proof of Theorem 1.1

Set $n := k - 2$ throughout. The argument is k -uniform on the $\dim S_k = 1$ family $k \in \{12, 16, 18, 20, 22, 26\}$: every step below — the cocycle structure of §2, the δ_T -solvability of Lemma 3.1, the Bernoulli closed-form inversion of Lemma 3.2, the Hurwitz-zeta kernel identification of Proposition 3.6, and the ℓ - and τ_0 -independence of Corollary 3.8 — holds for every even k with the Brown cuspidal-basis coefficient $c_S^{\ell-1}$ replaced by the relevant weight's basis-polynomial coefficient. The lower even weights $k \in \{2, 4, 6, 8, 10, 14\}$ in the $\dim S_k \leq 1$ family have $S_k = 0$, so the theorem is vacuous there. Numerical verification at $k = 12$ against Brown's published values for the canonical pair $f \in \{\Delta, \hat{\Delta}\}$ is consolidated in §3.1.

Lemma 3.1 (Solvability of $\delta_T P = C_T^f$ is cuspidality of f). *Define the linear endomorphism $\delta_T: V_n \rightarrow V_n$ by $\delta_T(P) := P|_T - P$, where $T \in \Gamma$ acts on V_n as $(X, Y) \mapsto (X + Y, Y)$. Then:*

- (a) $\ker(\delta_T) = \mathbb{C} \cdot Y^n$.
- (b) $\text{im}(\delta_T) = \{Q \in V_n : [Q]_{X^n} = 0\}$.

(c) For any meromorphic modular form f of weight $k = n + 2$ and any $\tau_0 \in \mathbb{H}$ admissible for f ,

$$[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} \cdot c_f(0), \quad (17)$$

where $c_f(0)$ is the constant Fourier coefficient of f .

Consequently, the equation $\delta_T P = C_T^f(\tau_0)$ is solvable for $P \in V_n$ if and only if $c_f(0) = 0$, i.e. if and only if f is a (weak) cusp form. When solvable, P is unique modulo $\mathbb{C} \cdot Y^n$.

Proof. (a) T acts on V_n as the translation $X \mapsto X + Y$, fixing only powers of Y . Hence T -invariants in V_n , equivalently $\ker(\delta_T)$, are exactly $\mathbb{C} \cdot Y^n$.

(b) Writing $P = \sum_{\ell=0}^n p_\ell X^{n-\ell} Y^\ell$ and expanding $(X + Y)^{n-\ell}$ binomially,

$$[\delta_T P]_{X^{n-m} Y^m} = \sum_{\ell=0}^{m-1} \binom{n-\ell}{m-\ell} p_\ell \quad (m = 0, 1, \dots, n). \quad (18)$$

This is strictly lower-triangular in the ordered basis $(X^n, X^{n-1}Y, \dots, Y^n)$: the $m = 0$ row is the empty sum, identically zero, and the off-diagonal entries $\binom{n-\ell}{m-\ell}$ for $\ell < m$ are positive. The $m = 0$ row forces $[\delta_T P]_{X^n} = 0$ for every P , while rows $m = 1, \dots, n$ determine $p_0, \dots, p_{n-1}$ uniquely from any prescribed right-hand side, with p_n unconstrained. Hence $\text{im}(\delta_T) = \{Q \in V_n : [Q]_{X^n} = 0\}$, and the solution P is unique modulo $\mathbb{C} \cdot Y^n$.

(c) The moment expansion (14) gives $[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} I_T^0(f, \tau_0)$ where $I_T^0(f, \tau_0) = \int_{\tau_0-1}^{\tau_0} f(\tau) d\tau$. Substituting the Fourier expansion $f(\tau) = \sum_n c_f(n) e^{2\pi i n \tau}$, absolutely convergent on the compact T -segment, and integrating term by term,

$$\int_{\tau_0-1}^{\tau_0} e^{2\pi i n \tau} d\tau = \begin{cases} 1 & \text{if } n = 0, \\ 0 & \text{if } n \neq 0, \end{cases}$$

by T -periodicity over a unit segment. Hence $I_T^0(f, \tau_0) = c_f(0)$ for every admissible τ_0 , which proves (17).

Combining (b) and (c) gives the solvability statement. $\square$

Lemma 3.2 (Bernoulli closed form for δ_T^{-1}). *For every $Q \in V_n$ with vanishing X^n -coefficient there exists $P \in V_n$ unique modulo $\mathbb{C} \cdot Y^n$ with $\delta_T P = Q$. Writing $Q = \sum_{r=1}^n q_r X^{n-r} Y^r$ and $P = \sum_{m=0}^n p_m X^{n-m} Y^m$, the solution is given in closed form by*

$$p_m = \sum_{r=1}^{m+1} \frac{B_{m-r+1} (n-r)!}{(m-r+1)! (n-m)!} q_r \quad (m = 0, 1, \dots, n-1), \quad (19)$$

where B_ℓ are Bernoulli numbers ($B_0 = 1$, $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$, $B_{2\ell+1} = 0$ for $\ell \geq 1$); the coefficient p_n is unconstrained and parametrises $\ker(\delta_T) = \mathbb{C} \cdot Y^n$.

Proof. The operator identity $T = e^{Y\partial_X}$ on V_n (since T acts as the translation $X \mapsto X + Y$ by Y) gives $\delta_T = e^{Y\partial_X} - 1$. The operator $Y\partial_X$ is nilpotent on V_n : each application reduces the X -degree by one, so $(Y\partial_X)^{n+1} = 0$ on V_n . Consequently, the formal power series in $Y\partial_X$ obtained from the Bernoulli generating function $z/(e^z - 1) = \sum_{k \geq 0} (B_k/k!) z^k$ truncates to a finite $\mathbb{Q}$ -linear combination of operators on V_n , and supplies the closed-form inverse

$$\delta_T^{-1} = (Y\partial_X)^{-1} \frac{Y\partial_X}{e^{Y\partial_X} - 1} = \sum_{k \geq 0} \frac{B_k}{k!} (Y\partial_X)^{k-1}, \quad (20)$$

in which the $k = 0$ summand is the formal antiderivative $(Y\partial_X)^{-1}$, acting on monomials by $(Y\partial_X)^{-1}(X^a Y^b) = X^{a+1} Y^b / (a+1)$ (for $b \geq 1$), and the $k \geq 1$ summands are differential operators that eventually annihilate any polynomial of bounded X -degree. Applied to $X^{n-r} Y^r$ for $r \geq 1$,

$$\delta_T^{-1}(X^{n-r} Y^r) = \sum_{k=0}^{n-r+1} \frac{B_k}{k!} \frac{(n-r)!}{(n-r-k+1)!} X^{n-r-k+1} Y^{r+k-1}. \quad (21)$$

Summing $Q = \sum_r q_r X^{n-r} Y^r$ termwise and reading off the $X^{n-m} Y^m$ coefficient identifies $k = m - r + 1$ (so r runs over $\{1, \dots, m+1\}$, k over $\{0, \dots, m\}$), yielding (19). Existence and uniqueness modulo $\mathbb{C} \cdot Y^n$ follow from Lemma 3.1(a, b). $\square$

Definition 3.3 (Bernoulli T -segment kernel $\widehat{k}_T(\ell, \tau)$). *For non-negative integers l with $l \leq n-1$, set the $\mathbb{Q}$ -rational coefficients*

$$\beta_{l,r} := \frac{(-1)^r \binom{n}{r} B_{l-r+1} (n-r)!}{(l-r+1)! (n-l)!} \quad \text{for } 1 \leq r \leq l+1, \quad (22)$$

and $\beta_{l,r} := 0$ for r outside $\{1, \dots, l+1\}$. For each interior $\ell \in \{2, \dots, k-2\}$, define the $\mathbb{Q}$ -rational polynomial

$$\widehat{k}_T(\ell, \tau) := \sum_{r=1}^n \left[\beta_{\ell-1,r} - (-1)^{\ell-1} \beta_{n-\ell+1,r} \right] \tau^r \in \mathbb{Q}[\tau], \quad (23)$$

of degree $\leq n$.

Lemma 3.4 (Integer- s collapse of the Hurwitz-zeta T -kernel). *For each integer $\ell \in \{2, \dots, k-2\}$, the Hurwitz-zeta T -kernel $\widetilde{k}_T(\tau, s)$ of (3) specialised to $s = \ell$ collapses to the $\mathbb{Q}$ -rational polynomial*

$$\widetilde{k}_T(\tau, \ell) = -\frac{B_\ell(\tau+1)}{\ell} + (-1)^{\ell-1} \frac{B_{k-\ell}(\tau+1)}{k-\ell} \in \mathbb{Q}[\tau] \quad (24)$$

of degree at most $k-2$.

Proof. The Hurwitz-zeta collapse $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$ for non-negative integer m , applied at $m = \ell-1$ and $m = k-\ell-1$ (both non-negative for $\ell \in \{2, \dots, k-2\}$), together with $e^{i\pi(\ell-1)} = (-1)^{\ell-1}$, converts (3) at $s = \ell$ to (24). $\square$

Lemma 3.5 (Bernoulli-form vs. Hurwitz-form T -kernel: polynomial identity and integral equivalence). *For each interior $\ell \in \{2, \dots, k-2\}$, the Bernoulli T -kernel $\widehat{k}_T(\ell, \tau)$ of (23) and the Hurwitz T -kernel $\widetilde{k}_T(\tau, \ell)$ of (24) are related by the polynomial identity*

$$\widehat{k}_T(\ell, \tau) = (-1)^{\ell-1} \binom{n}{\ell-1} \left[\widetilde{k}_T(\tau, \ell) - \widetilde{k}_T(0, \ell) \right] \in \mathbb{Q}[\tau]. \quad (25)$$

Equivalently, the difference $\widehat{k}_T(\ell, \tau) - (-1)^{\ell-1} \binom{n}{\ell-1} \widetilde{k}_T(\tau, \ell) = -(-1)^{\ell-1} \binom{n}{\ell-1} \widetilde{k}_T(0, \ell) \in \mathbb{Q}$ is a constant. Consequently, for every $f \in S_k^!$ and every $\tau_0 \in \mathbb{H}$,

$$\int_{\gamma_{\tau_0}^T} f(\tau) \widehat{k}_T(\ell, \tau) d\tau = (-1)^{\ell-1} \binom{n}{\ell-1} \int_{\gamma_{\tau_0}^T} f(\tau) \widetilde{k}_T(\tau, \ell) d\tau. \quad (26)$$

Proof. Polynomial identity. Both sides of (25) are $\mathbb{Q}$ -rational polynomials in τ of degree $\leq k-2$, and both have vanishing constant term: the left side by construction ((23) sums from $r=1$), and the right side because $\tilde{k}_T(\tau, \ell) - \tilde{k}_T(0, \ell)$ vanishes at $\tau=0$. By direct comparison of the τ^r coefficients for $1 \leq r \leq k-2$ using (23) for the left side and (24) for the right side, together with the standard Bernoulli polynomial expansion $B_n(\tau+1) = \sum_{s=0}^n \binom{n}{s} B_s \tau^{n-s} + n \tau^{n-1}$, the two polynomial coefficients agree. Hence the polynomial identity (25) holds in $\mathbb{Q}[\tau]$.

Integral equivalence. The difference $\tilde{k}_T(\ell, \tau) - (-1)^{\ell-1} \binom{n}{\ell-1} \tilde{k}_T(\tau, \ell) = -(-1)^{\ell-1} \binom{n}{\ell-1} \tilde{k}_T(0, \ell) \in \mathbb{Q}$ is constant by (25). For $f \in S_k^!$ the integral of any constant over the T -segment vanishes by cuspidality: $\int_{\gamma_{\tau_0}^T} f(\tau) d\tau = c_f(0) = 0$ (Lemma 3.1(c)). Hence the constant difference contributes zero and (26) follows. $\square$

Proposition 3.6 (Integral representation of r_f at the interior monomial). *Let $f \in S_k^!$ and $\tau_0 \in \mathbb{H}$. Let $P \in V_n/\mathbb{C}Y^n$ be the unique solution of $\delta_T P = (2\pi i)^{-(n+1)} C_T^f(\tau_0)$ from Lemma 3.1 and Lemma 3.2, and define the modified-cocycle S -value*

$$r_f := C_S^f(\tau_0) - (2\pi i)^{n+1} (P|_S - P) \in V_n \otimes \mathbb{C}. \quad (27)$$

For each interior $\ell \in \{2, \dots, k-2\}$, the coefficient of the monomial $X^{n-(\ell-1)} Y^{\ell-1}$ in r_f admits the integral representation

$$[r_f]_{X^{n-(\ell-1)} Y^{\ell-1}} = (2\pi i)^{n+1} (-1)^{\ell-1} \binom{n}{\ell-1} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, \ell) d\tau \right], \quad (28)$$

with $\tilde{k}_T(\tau, s)$ the Hurwitz-zeta kernel of (3).

Proof. S-piece. The moment expansion (14) gives

$$C_S^f(\tau_0) = (2\pi i)^{n+1} \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell I_S^\ell(f, \tau_0) X^{n-\ell} Y^\ell,$$

so $[C_S^f(\tau_0)]_{X^{n-(\ell-1)} Y^{\ell-1}} = (2\pi i)^{n+1} \binom{n}{\ell-1} (-1)^{\ell-1} I_S^{\ell-1}(f, \tau_0)$, with $I_S^{\ell-1} = \int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau$; this contributes the S -segment piece of (28) against the bare Mellin kernel $\tau^{\ell-1}$.

T-piece, Bernoulli form. S acts on V_n as $(X, Y) \mapsto (-Y, X)$, so for $P = \sum_l p_l X^{n-l} Y^l$ one has $[P|_S - P]_{X^{n-(\ell-1)} Y^{\ell-1}} = (-1)^{\ell-1} p_{n-\ell+1} - p_{\ell-1}$ (n even). By Lemma 3.2, each p_l ($0 \leq l \leq n-1$) is the explicit $\mathbb{Q}$ -linear combination $p_l = \sum_{r=1}^{l+1} \beta_{l,r} I_T^r(f, \tau_0)$ with $\beta_{l,r}$ from (22) of Definition 3.3, obtained by substituting the rescaled T -moments $q_r = \binom{n}{r} (-1)^r I_T^r$ of $(2\pi i)^{-(n+1)} C_T^f(\tau_0)$ into (19). In terms of $\hat{k}_T(\ell, \tau)$ of (23), the T -segment piece of $[r_f]_{X^{n-(\ell-1)} Y^{\ell-1}}$ reads $(2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau) \hat{k}_T(\ell, \tau) d\tau$.

T-piece, Hurwitz form. By Lemma 3.5, (26), the Bernoulli T -integral of the preceding paragraph equals $(-1)^{\ell-1} \binom{n}{\ell-1} \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, \ell) d\tau$. Combined with the S -piece, this yields (28) with the bare Hurwitz kernel $\tilde{k}_T(\tau, \ell)$ inside. $\square$

Lemma 3.7 (Brown-basis extraction of $\omega^\pm(f)$). *Let $r_f \in V_n \otimes \mathbb{C}$ be the modified-cocycle S -value of (27). The Eichler–Shimura / Brown cuspidal-basis decomposition ([1, 13]) reads*

$$r_f = \omega^+(f) p_f^+ + \omega^-(f) p_f^- + (\text{Eisenstein-direction term}), \quad (29)$$

with the Eisenstein-direction term contributing only to the boundary monomials X^n, Y^n of r_f (as it is a multiple of the coboundary $Y^n|_S - Y^n = X^n - Y^n$; see Appendix B for the distinction between this

Eisenstein-direction contribution to r_f and the period polynomial of an actual Eisenstein series). Combined with the integral representation (28) of Proposition 3.6, this yields

$$\omega^\pm(f) = \frac{(2\pi i)^{k-1} (-1)^{\ell-1} \binom{k-2}{\ell-1}}{c_S^{\ell-1}} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, \ell) d\tau \right] \quad (30)$$

for every interior $\ell \in \{2, \dots, k-2\}$, with parity $\pm$ determined by $\ell-1 \bmod 2$ and Brown basis coefficient $c_S^{\ell-1} := [p_f^\pm]_{X^{n-(\ell-1)}Y^{\ell-1}} \in \mathbb{Q}$. Identity (30) is identity (4) of Theorem 1.1.

Proof. For interior $\ell-1 \in \{1, \dots, n-1\}$, the Eisenstein-direction term in (29) contributes zero at the monomial $X^{n-(\ell-1)}Y^{\ell-1}$, so $[r_f]_{X^{n-(\ell-1)}Y^{\ell-1}} = \omega^\pm(f) c_S^{\ell-1}$. Substituting (28) and dividing through by $c_S^{\ell-1}$ yields (30). $\square$

Corollary 3.8 (Basepoint and index independence). *The two-segment linear functional of Theorem 1.1, $\omega^\pm(f) = \frac{(2\pi i)^{k-1} (-1)^{\ell-1} \binom{k-2}{\ell-1}}{c_S^{\ell-1}} [\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, \ell) d\tau]$ of (4), is independent of $\tau_0 \in \mathbb{H}$ and of the interior index $\ell \in \{2, \dots, k-2\}$ within its parity sector. At each k with $\dim S_k = 1$, this yields a $(k-3)$ -fold redundant extraction of $\omega^\pm(f)$ from the Fourier coefficients of f , reflecting the 1-dimensionality of the cuspidal direction in $H^1(\Gamma; V_{k-2})$.*

Proof. τ_0 -independence. By (27), namely $r_f := C_S^f(\tau_0) - (2\pi i)^{n+1}(P|_S - P)$, r_f depends on τ_0 only through the cohomology class $[C^f] \in H^1(\Gamma; V_n \otimes \mathbb{C})$ and the choice of coboundary representative P . The class is τ_0 -invariant by the basepoint-shift coboundary identity $C_\gamma^f(\tau_0') - C_\gamma^f(\tau_0) = P|_\gamma - P$ for the antiderivative $P = (2\pi i)^{n+1}(F(\tau_0') - F(\tau_0))$ of (16) (valid for f without interior poles), and the coboundary representative $P(\tau_0)$ from Lemma 3.2 is the unique-modulo-CY n solution of $\delta_T P = (2\pi i)^{-(n+1)} C_T^f$. The residual CY n ambiguity lies in the Eisenstein direction: $Y^n|_S - Y^n = X^n - Y^n$ is a coboundary that contributes only to the boundary monomials X^n, Y^n of r_f . Therefore the cuspidal coordinates of r_f — the $\omega^\pm(f)$ — are τ_0 -invariant.

ℓ -independence. By the Eichler–Shimura decomposition and Brown’s basis ([1, 13]), r_f projects onto the 1-dimensional cuspidal direction as $\omega^+(f) p_f^+ + \omega^-(f) p_f^-$ modulo Eisenstein contributions. Reading off the coefficient of any interior monomial $X^{n-(\ell-1)}Y^{\ell-1}$ for $\ell \in \{2, \dots, k-2\}$ returns $\omega^\pm(f) c_S^{\ell-1}$ with $c_S^{\ell-1} := [p_f^\pm]_{X^{n-(\ell-1)}Y^{\ell-1}}$ the basis-polynomial coefficient. The prefactor $(2\pi i)^{k-1} (-1)^{\ell-1} \binom{k-2}{\ell-1} / c_S^{\ell-1}$ in (4) is exactly the basis-conversion factor together with the moment-expansion combinatorial structure that normalises the right side to $\omega^\pm(f)$ independently of ℓ .

At $k = 12$ this gives a 9-fold redundancy ($\ell \in \{2, \dots, 10\}$); at general k with $\dim S_k = 1$, the interior range $\{2, \dots, k-2\}$ has $k-3$ elements. $\square$

With Proposition 3.6 (integral representation of $[r_f]$ at interior monomials), Lemma 3.4 (integer- s collapse of $\tilde{k}_T$), Lemma 3.7 (Brown-basis extraction of $\omega^\pm$), and Corollary 3.8 (ℓ - and τ_0 -independence) in hand, we assemble the proof of Theorem 1.1. For reference, the theorem statement is:

Theorem 3.9 (Periods and quasi-periods as an explicit linear functional, $\dim S_k \leq 1$ – Theorem 1.1, restated). *Let $k \in 2\mathbb{Z}$ with $\dim S_k \leq 1$, and let $f \in S_k^!$. For the substantive cases $k \in \{12, 16, 18, 20, 22, 26\}$ where $\dim S_k = 1$, the period and quasi-period coefficients $\omega^\pm(f)$ are given by the explicit two-segment linear functional*

$$\omega^\pm(f) = \frac{(2\pi i)^{k-1} (-1)^{\ell-1} \binom{k-2}{\ell-1}}{c_S^{\ell-1}} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, \ell) d\tau \right] \quad (31)$$

for every $f \in S_k^!$, every $\tau_0 \in \mathbb{H}$, and every interior $\ell \in \{2, \dots, k-2\}$, with parity $\pm$ determined by $\ell - 1 \bmod 2$. Here $c_S^{\ell-1} := [p_f^\pm]_{X^{n-(\ell-1)}Y^{\ell-1}} \in \mathbb{Q}$ is the Brown cuspidal-basis coefficient at the matched monomial ([1, 13]), $\tilde{k}_T(\tau, s)$ is the Hurwitz-zeta T -kernel of (3), and its integer- $s = \ell$ specialisation collapses via $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$ ($m \geq 0$) to the $\mathbb{Q}$ -rational polynomial

$$\tilde{k}_T(\tau, \ell) = -\frac{B_\ell(\tau+1)}{\ell} + (-1)^{\ell-1} \frac{B_{k-\ell}(\tau+1)}{k-\ell} \in \mathbb{Q}[\tau] \quad (32)$$

of degree at most $k-2$. The integral for $\omega^\pm(f)$ in (31) depends neither on τ_0 nor on $\ell \equiv \ell' \pmod{2}$.

Proof of Theorem 1.1, aka Theorem 3.9. The proof proceeds in seven steps. Steps 1–3 motivate the operator δ_T as the natural object whose inversion delivers the cuspidal cocycle representative. Step 4 (Lemma 3.2) performs the inversion in closed form. Steps 5–7 read $\omega^\pm(f)$ off the resulting period polynomial.

Step 1: the canonical cuspidal representative has $C_T^f = 0$. Within the cohomology class $[C^f] \in H^1(\Gamma; V_n)$, consider the representative with vanishing T -component. When $C_T = 0$, the cocycle relations $C_{S^2} = 0$ and $C_{(ST)^3} = 0$ collapse onto C_S alone, and the surviving relations on C_S are exactly the two Eichler–Shimura period-polynomial relations $r_f|(1+S) = 0$ and $r_f|(1+U+U^2) = 0$ with $U = TS$. The $C_T = 0$ representative therefore has its S -value in the Eichler–Shimura period-polynomial space, where Brown’s basis $\{p_f^+, p_f^-, \text{Eisenstein}\}$ is defined and the cuspidal coordinates are $\omega^\pm(f)$. The proof below kills C_T^f by a coboundary modification.

Step 2: kill C_T^f by a coboundary modification. The only freedom inside the cohomology class is to add a coboundary $\delta P(\gamma) := P|_\gamma - P$ for some $P \in V_n$. Under this modification, $C_T^f(\tau_0) \mapsto C_T^f(\tau_0) - (P|_T - P)$. Killing C_T^f becomes the equation

$$\delta_T P := P|_T - P = C_T^f(\tau_0), \quad (33)$$

to be solved for $P \in V_n$. This identifies the linear endomorphism $\delta_T: V_n \rightarrow V_n$ of Lemma 3.1 as the precise object whose inversion delivers the desired coboundary representative.

Step 3: solvability of $\delta_T P = C_T^f$ is exactly cuspidality of f . Lemma 3.1 computes $[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} c_f(0)$ and shows that $\text{im}(\delta_T) = \{Q \in V_n : [Q]_{X^n} = 0\}$. Solvability of (33) is therefore equivalent to $c_f(0) = 0$, i.e. to f being a (weak) cusp form. The cuspidality hypothesis $f \in S_k^!$ is not a separate input to the construction: it is the solvability condition of the coboundary equation in Step 2.

Step 4: closed-form Bernoulli inversion of δ_T . The operator identity $T = e^{Y\partial_X}$ (since T acts on V_n as the translation $X \mapsto X+Y$ by Y) gives $\delta_T = e^{Y\partial_X} - 1$, and the Bernoulli generating function $z/(e^z - 1) = \sum_{k \geq 0} B_k z^k/k!$ supplies δ_T^{-1} as a finite operator series on V_n (terminating because ∂_X annihilates polynomials of bounded X -degree after finitely many applications). Lemma 3.2 records the explicit closed-form expression for $P \in V_n/\mathbb{C}Y^n$ in terms of Bernoulli numbers.

Step 5: the modified cocycle and its S -component r_f . With $P := \delta_T^{-1}((2\pi i)^{-(n+1)} C_T^f(\tau_0)) \in V_n/\mathbb{C}Y^n$ from Step 4, the modified cocycle $C'^f := C^f - (2\pi i)^{n+1} \delta P$ has $C_T'^f = 0$ by construction, and its S -value $r_f := C_S'^f(\tau_0) = C_S^f(\tau_0) - (2\pi i)^{n+1} (P|_S - P)$ of (27) is the Eichler–Shimura period polynomial of f . By the Eichler–Shimura / Brown cuspidal-basis decomposition (29) of Lemma 3.7, $\omega^\pm(f)$ are the explicit linear coordinates of r_f along $p_f^\pm$ in the 1-dimensional cuspidal direction.

Step 6: unpacking r_f into bare Mellin and Hurwitz integrals. Substituting the moment expansion of $C_S^f(\tau_0)$ from (14) and the Bernoulli expression for P from Step 4 into (27), and reading off the coefficient of $X^{n-(\ell-1)}Y^{\ell-1}$ for each interior $\ell \in \{2, \dots, k-2\}$, Proposition 3.6 produces the integral representation (28) of $[r_f]_{X^{n-(\ell-1)}Y^{\ell-1}}$ with the bare Mellin kernel $\tau^{\ell-1}$ on the S -segment and the bare Hurwitz kernel $\tilde{k}_T(\tau, \ell)$ on the T -segment. Lemma 3.4 (the integer- s collapse of $\tilde{k}_T$) exhibits the T -kernel as a $\mathbb{Q}$ -rational polynomial of degree $\leq k-2$. Lemma 3.7 (Brown-basis extraction) then divides through by $c_S^{\ell-1}$ to give (4).

Step 7: ℓ - and τ_0 -independence. Both follow from r_f being a single fixed polynomial in V_n that lies in the 1-dimensional cuspidal direction. Reading off any interior monomial coefficient of r_f returns a constant multiple of the same $\omega^\pm(f)$ (parity-selected by ℓ), and the prefactor $(-1)^{\ell-1} \binom{k-2}{\ell-1} (2\pi i)^{k-1} / c_S^{\ell-1}$ in (4) normalises away the basis-conversion factor. Independence in τ_0 follows from $[C^f]$ being a cohomology-class invariant; the residual $\mathbb{C}Y^n$ ambiguity in the coboundary representative P contributes only to the Eisenstein direction. Corollary 3.8 makes both statements precise; at $k=12$, the 9 interior values of ℓ give the 9-fold redundant extraction. $\square$

3.1 Numerical verification

At $k=12$, every step of the period-extraction pipeline of Theorem 1.1 has been verified against Brown’s published values [1] for the canonical pair $f \in \{\Delta, \hat{\Delta}\}$ in the companion script `verify_periods.py` [4], run at 50-digit `mpmath` precision.

	$\Delta \in S_{12}$, period $\omega^\pm(\Delta)$	$\hat{\Delta} \in S_{12}^!$, quasi-period $\eta^\pm(\hat{\Delta})$
+	-68 916 772.80959519...	127 202 100 647.17709...
-/i	- 5 585 015.37931040...	10 276 732 343.64913...

Table 1: Periods of Δ and quasi-periods of $\hat{\Delta}$ at $k=12$, reproduced from the Fourier coefficients of f by (4) to relative residual $\lesssim 10^{-42}$ at $\tau_0 = 0.3 + 1.2i$. Reference values from [1].

The verification is layered through the proof of Theorem 1.1 as follows; the relevant named functions in `verify_periods.py` [4] are indicated in `typewriter`.

- **Cocycle integration** ((12), (14)): `cocycle_poly` computes $C_S^f(\tau_0), C_T^f(\tau_0) \in V_{10} \otimes \mathbb{C}$ by V_{10} -valued contour integration along $\gamma_{\tau_0}^{S,T}$.
- **Coboundary inversion two ways** (Lemma 3.2): `solve_coboundary_LU` solves $\delta_T P = (2\pi i)^{-(n+1)} C_T^f$ by forward substitution on the strictly-lower-triangular matrix (18); `solve_coboundary_bernoulli` uses the closed form (19) of Lemma 3.2. The two outputs agree to relative residual $\sim 10^{-40}$, confirming (19).
- **Brown-basis extraction at 9 interior monomials** (Lemma 3.7): `extract_periods` reads off $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$ from $r_f = C_S^f - (2\pi i)^{n+1}(P|_S - P)$ at each interior $\ell \in \{2, \dots, 10\}$. All 9 extractions agree pairwise — the explicit ℓ -independence of Corollary 3.8 — to $\lesssim 10^{-42}$, matching the Brown reference values in Table 1.
- **Basepoint independence** (Corollary 3.8): re-extraction of $\eta^\pm(\hat{\Delta})$ at $\tau'_0 = -0.2 + 0.9i$ agrees with the $\tau_0 = 0.3 + 1.2i$ values to $\lesssim 10^{-42}$.

- **Clean-form identity** (4) (Theorem 1.1): `hurwitz_extract` evaluates the bare-Mellin + bare-Hurwitz integral form directly at $\ell \in \{2, 3, 4, 5\}$ for $f = \Delta$, agreeing with the Brown reference values in Table 1 to $\lesssim 10^{-42}$.
- **S - vs T -segment growth-rate cancellation** for $f = \hat{\Delta}$: `bare_integrals` confirms that $|I_S|, |I_T|$ each grow $\sim 10^7$ -fold across $\text{Im}(\tau_0) \in [1.2, 2.5]$ while $I_S + I_T$ stays τ_0 -invariant.

Analogous verification at higher weights $k \in \{16, 18, 20, 22, 26\}$ where $\dim S_k = 1$ requires only the Brown basis coefficients $c_S^{\ell-1}$ at those weights and is structurally identical.

4 The Hurwitz-zeta extension and the L -function on $M_k^!$: proof of Theorem 1.2

4.1 Derivation of the kernel $\tilde{k}_T$

We seek a kernel $\tilde{k}_T(\tau, s)$, analytic in $\tau \in \mathbb{H}$ and meromorphic in $s \in \mathbb{C}$, such that $L^*(f, s; \tau_0)$ as defined in (6) is independent of τ_0 for every $f \in M_k^!$. Differentiating (6) in τ_0 and using $\Delta(-1/\tau_0) = \tau_0^k \Delta(\tau_0)$ (the S -modularity of f at the S -segment endpoint $-1/\tau_0$) together with $f(\tau_0 - 1) = f(\tau_0)$ (the T -modularity at $\tau_0 - 1$) gives, after cancelling the common factor $i^{-s} f(\tau_0)$,

$$\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s) = -\tau_0^{s-1} + e^{i\pi(s-1)} \tau_0^{k-1-s}. \quad (34)$$

The right side picks up the S -segment endpoint contribution at $-1/\tau_0 = e^{i\pi} \tau_0^{-1}/(-1) = e^{i\pi}/\tau_0$ via the branch identity $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)} \tau_0^{-(s-1)}$ combined with the Jacobian $d(-1/\tau_0)/d\tau_0 = \tau_0^{-2}$ and the modularity factor τ_0^k , giving $e^{i\pi(s-1)} \tau_0^{k-1-s}$ after simplification.

Equation (34) is the complex- s analog of the polynomial coboundary equation $\delta_T P = Q$ of §3. The Hurwitz raising identity

$$\zeta(\sigma, a+1) - \zeta(\sigma, a) = -a^{-\sigma}$$

inverts the shift operator $\tau \mapsto \tau - 1$ for any complex σ , on the analytic function $\zeta(\sigma, \tau+1)$. Applying at $(\sigma, a) = (1-s, \tau_0)$ and $(\sigma, a) = (s-(k-1), \tau_0)$, and combining with the sign $e^{i\pi(s-1)}$ gives (3).

Lemma 4.1. *For $f \in M_k^!$, $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ defined by (6) is independent of $\tau_0 \in \mathbb{H}$ for every $s \in \mathbb{C}$.*

Proof. Direct τ_0 -differentiation, using (34) to cancel the boundary contributions from the two segments. Both integrals in (6) are finite at every $\tau_0 \in \mathbb{H}$ because f has no poles in $\mathbb{H}$ and the kernels $\tau^{s-1}, \tilde{k}_T(\tau, s)$ are analytic in $\tau \in \mathbb{H}$ for every fixed $s \in \mathbb{C}$ (the Hurwitz $\zeta(\sigma, a)$ is analytic in a on $\text{Re}(a) > 0$). $\square$

4.2 The Hurwitz-unfolding lemma

Lemma 4.2 (Hurwitz unfolding). *For every positive integer n and every $\nu \in \mathbb{C}$,*

$$\int_i^{i+1} e^{2\pi i n \sigma} \zeta(\nu, \sigma) d\sigma = i^{1-\nu} (2\pi n)^{\nu-1} \Gamma(1-\nu, 2\pi n). \quad (35)$$

Proof. First assume $\text{Re}(\nu) > 1$, where the Hurwitz series $\zeta(\nu, \sigma) = \sum_{m \geq 0} (\sigma + m)^{-\nu}$ converges uniformly on the compact segment $[i, i+1]$ [6, §25.11]. Series-integral exchange and the substitution

$u = \sigma + m$ in the m -th summand collapse the sum to a single integral by integer- n periodicity $e^{2\pi inm} = 1$:

$$\int_i^{i+1} e^{2\pi in\sigma} \zeta(\nu, \sigma) d\sigma = \sum_{m \geq 0} \int_{i+m}^{i+m+1} e^{2\pi inu} u^{-\nu} du = \int_i^{i+\infty} e^{2\pi inu} u^{-\nu} du,$$

an integral along the horizontal ray $\{i + t : t \geq 0\} \subset \mathbb{H}$.

Substitute $t = -2\pi inu$: $u = it/(2\pi n)$, $du = i dt/(2\pi n)$, and $u^{-\nu} = i^{-\nu}(2\pi n)^{\nu} t^{-\nu}$ on the principal branch (consistent with $\arg(u) = \pi/2$ at $u = i$ and $\arg(t) = 0$ at $t = 2\pi n$). The contour in t runs from $2\pi n$ down to $2\pi n - i\infty$, in the right half-plane. Since $e^{-t} t^{-\nu}$ decays exponentially as $|t| \rightarrow \infty$ in $\text{Re}(t) > 0$, Cauchy's theorem applied to the rectangle $\{2\pi n \leq \text{Re}(t) \leq R, -R \leq \text{Im}(t) \leq 0\}$ in the limit $R \rightarrow \infty$ deforms the vertical contour $[2\pi n, 2\pi n - i\infty]$ to the real-axis contour $[2\pi n, +\infty]$:

$$\int_{2\pi n}^{2\pi n - i\infty} e^{-t} t^{-\nu} dt = \int_{2\pi n}^{\infty} e^{-t} t^{-\nu} dt = \Gamma(1 - \nu, 2\pi n)$$

by the integral definition of the upper incomplete gamma [6, §8.2]. Collecting prefactors gives (35) for $\text{Re}(\nu) > 1$.

Both sides of (35) are entire functions of ν : the upper incomplete gamma $\Gamma(s, x)$ is entire in s for fixed $x \neq 0$ [6, §8.2.7], and the integral on the left is entire in ν because $\zeta(\sigma, \sigma_0)$ is meromorphic in σ with a simple pole only at $\sigma = 1$, whose residue contribution to $\int_{i-1}^i e^{2\pi in\tau} d\tau = 0$ for nonzero integer n (since $e^{2\pi in(i-1)} = e^{2\pi in \cdot i}$). Analytic continuation in ν extends (35) to all complex ν . $\square$

4.3 Per-mode evaluation: proof of Theorem 1.2

By Lemma 4.1, the value $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = L^*(f, s)$ is common to all admissible contours and basepoints. At the elliptic fixed-point basepoint $\tau_0 = i$, the S -segment $[-1/i, i] = [i, i]$ has zero length and the S -integral in (6) vanishes identically, leaving

$$L^*(f, s; \gamma_{\tau_0}^S(i) = 0, \gamma_{\tau_0}^T(i)) = i^{-s} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau.$$

Expanding $f = \sum_n c_f(n) q^n$ as a q -series and swapping sum and integral (justified by uniform convergence of the q -series on the compact segment $[i-1, i]$, including any finite number of polar modes since $|q^{-1}| = e^{2\pi}$ is bounded there) gives

$$L^*(f, s) = i^{-s} \sum_n c_f(n) \int_{i-1}^i e^{2\pi in\tau} \tilde{k}_T(\tau, s) d\tau. \quad (36)$$

We treat the constant Fourier mode $n = 0$ and the non-constant modes $n \neq 0$ separately.

The constant Fourier mode. For $n = 0$, the integrand reduces to $\tilde{k}_T(\tau, s)$ itself. The Hurwitz zeta has antiderivative $\int \zeta(\sigma, a) da = -\zeta(\sigma - 1, a)/(\sigma - 1) + C$ for $\sigma \neq 1$, and the Hurwitz raising identity $\zeta(\sigma - 1, a + 1) - \zeta(\sigma - 1, a) = -a^{1-\sigma}$ at $a = i$ gives

$$\int_{i-1}^i \zeta(\sigma, \tau + 1) d\tau = \int_i^{i+1} \zeta(\sigma, u) du = \frac{i^{1-\sigma}}{\sigma - 1}. \quad (37)$$

Applying (37) to each half of $\tilde{k}_T$ at $\sigma = 1 - s$ and $\sigma = s - (k - 1)$:

$$\int_{i-1}^i \tilde{k}_T(\tau, s) d\tau = -\frac{i^s}{s} + e^{i\pi(s-1)} \frac{i^{k-s}}{k-s}.$$

The overall i^{-s} prefactor and the phase collapse $i^{-s} e^{i\pi(s-1)} i^{k-s} = -i^k$ give the constant-term contribution

$$c_f(0) \cdot i^{-s} \int_{i-1}^i \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right). \quad (38)$$

This matches the polar term of [3, Lemma 2.1] (equivalently, the polar piece of [2, Thm. 1.1]) at the splitting point $B = 1$.

The non-constant Fourier modes. For $n \neq 0$, substituting $\sigma = \tau + 1$ shifts the contour to $[i, i+1]$ with the prefactor $e^{-2\pi in} = 1$. Apply Lemma 4.2 to each half of $\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1)$, i.e. at $\nu = 1-s$ and $\nu = s-(k-1)$:

$$\begin{aligned} \int_i^{i+1} e^{2\pi in\sigma} \zeta(1-s, \sigma) d\sigma &= i^s (2\pi n)^{-s} \Gamma(s, 2\pi n), \\ \int_i^{i+1} e^{2\pi in\sigma} \zeta(s-(k-1), \sigma) d\sigma &= i^{k-s} (2\pi n)^{s-k} \Gamma(k-s, 2\pi n). \end{aligned}$$

The first half against the i^{-s} prefactor gives $\Gamma(s, 2\pi n)/(2\pi n)^s$; the second half against the relative phase $-e^{i\pi(s-1)}$ collapses via $-i^{-s} e^{i\pi(s-1)} i^{k-s} = i^k$ to $i^k \Gamma(k-s, 2\pi n)/(2\pi n)^{k-s}$. Adding the two halves gives the per-mode summand

$$c_f(n) \cdot i^{-s} \int_{i-1}^i e^{2\pi in\tau} \tilde{k}_T(\tau, s) d\tau = c_f(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right], \quad (39)$$

which is the $B = 1$ value of the cuspidal summand of [3, Lemma 2.1].

Summing (38) and (39) over all Fourier modes reproduces the closed-form sum (7).

4.4 Worked example at $k = 12$: Bernoulli polynomials and numerics

At $k = 12$ the kernel of (3) is

$$\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-11, \tau+1).$$

At integer values of $s \in \{1, 2, \dots, 11\}$, both Hurwitz zeta factors collapse to Bernoulli polynomials via $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$ ($m \geq 0$), and $\tilde{k}_T(\tau, s)$ becomes a polynomial of degree ≤ 11 in τ . Three representative values:

$$\begin{aligned} \tilde{k}_T(\tau, 1) &= -\tau - \frac{1}{2} + \frac{1}{11} B_{11}(\tau+1), & \tilde{k}_T(\tau, 6) &= -\frac{1}{3} B_6(\tau+1), \\ \tilde{k}_T(\tau, 11) &= -\frac{1}{11} B_{11}(\tau+1) + \tau + \frac{1}{2}. \end{aligned}$$

The $s \leftrightarrow 12-s$ symmetry follows from the $s \leftrightarrow k-s$ structure of the two-term kernel (with $i^k = +1$ at $k = 12$).

Identification with Theorem 1.1. With the clean form of Theorem 1.1, the T -segment kernel of (4) is exactly the integer- $s = \ell$ specialisation $\tilde{k}_T(\tau, \ell)$ of the Hurwitz-zeta kernel of (3), with no convention conversion required: all combinatorial structure $(-1)^{\ell-1} \binom{k-2}{\ell-1} / c_S^{\ell-1}$ governing the $\omega^\pm(f)$ normalisation lives outside the two-segment bracket. The period-extraction functional (4) is thus the rescaled integer- s shadow of the all- s L -function functional (6) of Theorem 1.2, with the bare Mellin τ^{s-1} and Hurwitz $\tilde{k}_T(\tau, s)$ kernels mapped to their integer- $s = \ell$ specialisations and the analytic prefactor i^{-s} replaced by the combinatorial Brown-basis factor.

Numerical values. Running the companion `mpmath` scripts [4] at 50-digit precision, with $\tau_0 \in \{0.3 + 1.2i, i, 1.5i, -0.4 + 0.8i\}$ and $f \in \{\Delta, \hat{\Delta}\}$, reproduces the values of $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$ of [1]:

$$\begin{aligned}\omega^+(\Delta) &\approx -68916772.8096\dots, & \omega^-(\Delta)/i &\approx -5585015.3793\dots, \\ \eta^+(\hat{\Delta}) &\approx 1.27202 \times 10^{11}, & \eta^-(\hat{\Delta})/i &\approx 1.02767 \times 10^{10},\end{aligned}$$

with relative residuals $\lesssim 10^{-40}$, and the right side is independent of $\ell \in \{2, \dots, 10\}$ and of the basepoint across all four τ_0 choices.

Complex- s verification. Beyond the integer- ℓ period extraction, the all- s functional (6) was checked directly against the canonical completed L -function $L^*(\Delta, s)$ of Theorem 1.2 at non-integer s [4]. At half-integer points $s \in \{1.5, 2.5, \dots, 6.5\}$, off-axis complex points $s \in \{2.7 + 0.4i, 5 + i, 5.5 + 2.5i\}$, and points outside the critical strip $s \in \{0.5, -0.3\}$, the finite-contour value agrees with $L^*(\Delta, s)$ to relative residual $\sim 10^{-53}$ at 50-digit precision, and is independent of $\tau_0 \in \{0.3 + 1.2i, 1.5i, -0.4 + 0.8i, 0.2 + 2i\}$ at the same level. The corresponding check for $\hat{\Delta}$ holds to $\sim 10^{-47}$, the loss of digits reflecting the q^{-1} cusp mode.

5 Wall-crossing for meromorphic forms: proof of Theorem 1.3

5.1 Existence and local constancy

Let f be a meromorphic modular form of even weight k on Γ , and let $\tau_0 \in \mathbb{H}$ be a basepoint such that the contour pair (1) avoids the Γ -orbit of $f^{-1}(\infty) \cap \mathbb{H}$. The integrands $f(\tau) \tau^{s-1}$ and $f(\tau) \tilde{k}_T(\tau, s)$ are continuous on each compact segment for every fixed $s \in \mathbb{C}$ (the kernels are analytic in $\tau \in \mathbb{H}$), so both integrals in (2) are finite and $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is well-defined. This proves Theorem 1.3(a).

For local constancy, the argument of §4.1 applies verbatim: differentiating (2) in τ_0 gives the boundary contributions

$$i^{-s} f(\tau_0) \left[\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s) + \tau_0^{s-1} - e^{i\pi(s-1)} \tau_0^{k-1-s} \right] = 0$$

by (34), provided the modularity identities $f(-1/\tau_0) = \tau_0^k f(\tau_0)$ and $f(\tau_0 - 1) = f(\tau_0)$ hold at the contour endpoints (which they do for every meromorphic modular form, irrespective of the location of interior poles). Hence $dL^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)/d\tau_0 = 0$ whenever the contour pair varies within the admissible open subset of $\mathbb{H}$. This establishes Theorem 1.3(b).

5.2 Wall-crossing across an interior pole

Suppose a continuous one-parameter deformation of τ_0 to τ'_0 in $\mathbb{H}$ moves the S -segment $[-1/\tau_0, \tau_0]$ across an interior pole $\tau_p \in \mathbb{H}$ of f (the T -segment case is identical up to the choice of kernel; see below). Concretely, let $\Sigma_t \subset \mathbb{H}$ denote the S -segment at parameter $t \in [0, 1]$, with Σ_0 the S -segment at τ_0 and Σ_1 at τ'_0 , and assume that the swept-out region $\bigcup_t \Sigma_t$ encloses τ_p with winding number $\sigma_p^S \in \{+1, -1\}$ (orientation of the crossing). By Cauchy's theorem applied to the closed contour $\Sigma_1 - \Sigma_0$,

$$\int_{\Sigma_1} f(\tau) \tau^{s-1} d\tau - \int_{\Sigma_0} f(\tau) \tau^{s-1} d\tau = 2\pi i \sigma_p^S \text{Res}(f, \tau_p) \tau_p^{s-1}$$

for a simple pole of f at τ_p . An identical argument applies when the T -segment $[\tau_0 - 1, \tau_0]$ sweeps across τ_p , yielding

$$\int_{T\text{-seg at } \tau'_0} - \int_{T\text{-seg at } \tau_0} = 2\pi i \sigma_p^T \text{Res}(f, \tau_p) \tilde{k}_T(\tau_p, s),$$

where $\sigma_p^T \in \{+1, -1\}$ encodes the orientation of the T -segment crossing. Combining (with the i^{-s} prefactor) gives

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) - L^*(f, s; \gamma_{\tau_0'}^S, \gamma_{\tau_0'}^T) = -2\pi i i^{-s} \text{Res}(f, \tau_p) \left[\sigma_p^S \tau_p^{s-1} + \sigma_p^T \tilde{k}_T(\tau_p, s) \right]. \quad (40)$$

This is the residue formula $\mathcal{R}_{\tau_p}(f, s)$ of (10), with the winding-number indices σ_p^S, σ_p^T making explicit the dependence on which segment(s) cross. At a generic wall only one of σ_p^S, σ_p^T is nonzero; both can be nonzero on the codimension-two loci where the S - and T -segments simultaneously cross Γ -images of τ_p .

For a pole of order $m \geq 1$ at τ_p with Laurent expansion $f(\tau) = \sum_{\ell \geq -m} r_{f, \tau_p}(\ell) (\tau - \tau_p)^\ell$, the residue formula (40) generalises by replacing $\text{Res}(f, \tau_p) K(\tau_p, s)$ with $\sum_{\ell=1}^m r_{f, \tau_p}(-\ell) \partial_{\tau_p}^{\ell-1} K(\tau_p, s) / (\ell-1)!$ for each relevant kernel $K \in \{\tau^{s-1}, \tilde{k}_T(\tau, s)\}$, by the standard residue formula for higher-order poles.

5.3 Reduction on regular-at-cusps meromorphic forms

Let $f \in F_k$ be regular at the cusp $i\infty$, so $f(\tau) = \sum_{n \geq N} c_f(n) q^n$ for some $N \geq 0$.

Lemma 5.1 (Cusp decay of the T -segment). *For $f \in F_k$ regular at the cusp with $f(\tau) = O(q^N)$ at $i\infty$ for $N \geq 1$, every $s \in \mathbb{C}$, and every $\varepsilon \in \mathbb{R}$ fixed,*

$$\int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau = O(|\tau_0|^{C(\text{Re } s)} e^{-2\pi N \text{Im}(\tau_0)}) \quad \text{as } \text{Im}(\tau_0) \rightarrow \infty, \text{Re}(\tau_0) = \varepsilon.$$

The exponent $C(\text{Re } s) = \max(0, 1/2 - \text{Re}(1-s)) + \max(0, 1/2 - \text{Re}(s - (k-1))) + \varepsilon_0$ for any $\varepsilon_0 > 0$ is controlled by the standard polynomial-growth bound for the Hurwitz zeta on vertical strips [6, §25.11.36]. When $N = 0$ (i.e. f has a nonzero constant term at $i\infty$), the T -segment is $O(|\tau_0|^{C(\text{Re } s)})$ but does not decay; in this case the constant-term Fourier mode must be split off and treated separately (see [3, Lemma 2.1] for the analogous polynomial correction in the deformed-Mellin construction).

Proof. On the segment $\tau \in [\tau_0 - 1, \tau_0]$ with $\text{Im}(\tau) = \text{Im}(\tau_0)$, the bound $|f(\tau)| \leq M e^{-2\pi N \text{Im}(\tau_0)}$ holds for some $M = M(f)$ depending only on f . Combining with the polynomial-in- τ growth of $\tilde{k}_T(\tau, s)$ from [6, §25.11.36] and integrating over the length-1 segment gives the stated bound. $\square$

To match the deformed-Mellin construction of $L_f^*(s)_{\text{int}}$ (see [3]), take $\tau_0 = \varepsilon + i\Lambda$ for some small $\varepsilon \in \mathbb{R}$ chosen so that $\text{Re}(\tau_0) = \varepsilon$ avoids all Γ -images of interior poles of f . As $\Lambda \rightarrow \infty$:

- The T -segment integral $\rightarrow 0$ by Lemma 5.1 (using $N \geq 1$; for $N = 0$ the analog argument with the constant-term subtraction $c_f(0)$ from [3] applies).
- The S -segment $[-1/\tau_0, \tau_0]$ approaches the $\Lambda \rightarrow \infty$ limit of the Λ -truncated Mellin contour along $\text{Re}(\tau) = \varepsilon$ (modulo orientation, which fixes the L_{int}^* convention).

By Lemma 4.1 (extended to meromorphic f within a fixed connected component, by §5.1 above), the left side is Λ -independent, so

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = \lim_{\Lambda \rightarrow \infty} L^*(f, s; \gamma_{\varepsilon+i\Lambda}^S, \gamma_{\varepsilon+i\Lambda}^T) = L_f^*(s)_{\text{int}, \gamma_\varepsilon},$$

where γ_ε is the deformed-Mellin contour along $\text{Re}(\tau) = \varepsilon$. This proves Corollary 1.4. $\square$

5.4 Wall-crossing of the period-polynomial cocycle

The wall-crossing residue formula (10) acts on the complex- s functional $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$. Specialised at integer $s = \ell \in \{2, \dots, k-2\}$ and assembled into the V_{k-2} moment expansion of §2, it gives the wall-crossing of the period-polynomial cocycle of f under contour deformation across an interior pole.

Corollary 5.2 (Period-polynomial wall-crossing residue). *Let f be a meromorphic modular form of weight k on Γ , and let $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T), (\gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T)$ be two admissible contour pairs related by a single crossing of an interior pole $\tau_p \in \mathbb{H}$ of f . Denote by $C_S^f(\gamma_{\tau_0}^S), C_T^f(\gamma_{\tau_0}^T) \in V_{k-2} \otimes \mathbb{C}$ the corresponding V_{k-2} -valued cocycle values from (12). For a simple pole $\text{Res}(f, \tau_p)$,*

$$C_S^f(\gamma_{\tau_0}^S) - C_S^f(\gamma_{\tau'_0}^S) = 2\pi i (2\pi i)^{k-2} \sigma_{\tau_p}^S \text{Res}(f, \tau_p) (X - \tau_p Y)^{k-2}, \quad (41)$$

and analogously for C_T^f with $\sigma_{\tau_p}^S$ replaced by $\sigma_{\tau_p}^T$. The integer- $s = \ell$ specialisation of (10) is the coefficient of $\binom{k-2}{\ell} (-1)^\ell X^{k-2-\ell} Y^\ell$ in (41) (combined for S and T). For $\dim S_k = 1$ and $f \in F_k$ meromorphic with interior poles, the period polynomial r_f extracted via Theorem 1.1 from $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ therefore depends on the homotopy class of the contour pair, with jump (41) across each wall.

Proof. The integrand $f(\tau)(X - \tau Y)^{k-2}$ of (12) has a simple pole at $\tau = \tau_p$ with residue $\text{Res}(f, \tau_p)(X - \tau_p Y)^{k-2}$, so a contour deformation crossing τ_p with winding $\sigma_{\tau_p} \in \{-1, +1\}$ contributes $2\pi i \sigma_{\tau_p} \text{Res}(f, \tau_p)(X - \tau_p Y)^{k-2}$ to the moment integral. Multiplying by the cocycle prefactor $(2\pi i)^{k-2}$ of (12) gives (41). Expanding $(X - \tau_p Y)^{k-2}$ in the monomial basis and matching coefficients of $X^{k-2-\ell} Y^\ell$ for each $\ell \in \{0, \dots, k-2\}$ reproduces (10) at integer $s = \ell$ (with $\tau_p^{\ell-1}$ arising from $X^{k-2-\ell} Y^\ell$ at $\ell \geq 1$ from the S -segment contribution, and analogously for the T -segment via $\tilde{k}_T(\tau_p, \ell)$ at integer ℓ). $\square$

5.5 Remark on the polylog-projection machinery

The polylog projection $\hat{P}_{\tau_p} : f \mapsto \sum_{m>0} r_{f, \tau_p}^*(m) \text{Li}_{1-m}(e(\tau - \tau_p))$ (see [3, §2.2]) subtracts the leading polar tails of f at each interior pole τ_p , ensuring that the Λ -truncated Mellin integral against f has a finite $\Lambda \rightarrow \infty$ limit (necessary in the deformed-Mellin construction because of the Hagedorn-rate Fourier coefficients $|c_f(n)| \sim n^{p-1} e^{2\pi n \text{Im}(\tau_p)}$ generated by interior poles [3, Prop. 1.1, 2.7]). In the finite-contour formulation (2) the $\Lambda \rightarrow \infty$ limit is not taken, so $\hat{P}_{\tau_p}$ is not required for the existence or local-constancy properties of L^* . The polylog-projection machinery does remain relevant, however, for the *closed-form per-mode evaluation* of L^* in incomplete gammas parallel to (7), where the Hagedorn-rate Fourier modes must be split off and dealt with separately before each mode contributes a clean incomplete-gamma term. The systematic interior-pole closed form parallel to Theorem 1.2 is left to the sequel.

6 Outlook

Wilson-loop / holonomy interpretation. The period-defining contour pair, projected to the modular curve $Y(1)$, descends to the two generators of the orbifold fundamental group $\pi_1^{\text{orb}}(Y(1)) \cong \mathbb{Z}/2 * \mathbb{Z}/3$. The polynomial coefficient system V_{k-2} is a flat local system over $Y(1)$, and the cocycle C_γ^f is the weighted holonomy of this flat connection along the loop γ , with f supplying the gauge-invariant weight. The cohomology class $[C^f] \in H^1(\Gamma; V_{k-2})$ is the analog of a gauge-invariant observable; τ_0 is the analog of a trivialisation frame; $\omega^\pm(f)$ are coordinates of $[C^f]$ in the cuspidal subspace, the analog of physical observables read off a Wilson-loop expectation. The period-defining

finite-contour functional is, in this dictionary, the modular-form analog of a Wilson-loop holonomy on the modular curve.

The on-shell methods analogy. The classical Mellin construction is “off-shell” in the same sense as a Lagrangian gauge-theory amplitude calculation: the intermediate integrand $f(\tau) \tau^{s-1} d\tau$ encodes gauge-redundant information (the basepoint τ_0 , the contour to the cusp), and the gauge-invariant observable $L^*(f, s)$ is reached only after a non-canonical limit. Theorem 1.2 is an “on-shell” reformulation in which the gauge-invariant observable is the value of a finite-contour functional at any admissible basepoint, with no limit and no regulator. The modular-form analog of the on-shell S-matrix programme [15, 16] — constructing $\omega^\pm$ and $L^*(f, s)$ from $\{c_f(n)\}$ via a kernel that does not pass through the cocycle / coboundary scaffolding — is partially answered by Theorem 1.1 on the period side and Theorem 1.2 on the L -function side, with the remaining question being whether the kernels themselves admit a construction independent of the δ_T^{-1} Bernoulli operator.

Regularisation as a symptom of bad framing. The disappearance of any regularisation step in (2) is an instance of a recurring methodological pattern: *when a divergence arises only after one has made a non-canonical upstream choice, finding the canonical formulation eliminates the divergence together with the regulator.* The same move underlies the on-shell amplitudes programme [15]. The pattern is not universal — chiral anomalies in even dimensions are a counterexample, with cohomologically robust “regulator” content invariant under reframings — but it is a useful prior, and the finite-contour formulation here is a particularly clean instance in which the canonical $L^*(f, s)$ falls out as the value of a well-defined linear functional at any admissible basepoint, with no auxiliary parameter to take to infinity and no contour to deform to a cusp.

The geometric content of the Mellin kernel. The polynomial coefficient system V_{k-2} underlying the period construction is the symmetric-power local system that gives the polynomial kernel $(X - \tau Y)^{k-2}$ of the period polynomial. Its flat connection on the modular curve is the natural geometric home of $\omega^\pm(f)$. The Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ of (3) extends the polynomial kernel to a one-parameter analytic family in s , but the geometric content of the bare Mellin kernel $f(\tau) \tau^{s-1} d\tau$ itself — as opposed to its specialisation at integer s to a polynomial-in- V_{k-2} kernel — is not visible in the construction of this paper. A natural conjecture is that $\tau^{s-1} d\tau$ is a section of an infinite-dimensional principal-series-type local system on $Y(1)$, in the same one-parameter family as the Lewis–Zagier period functions for Maass wave forms [14]. Identifying the relevant local system explicitly, and connecting the Hurwitz-zeta inversion of δ_T to a coboundary statement in this larger system, is an open question.

Poincaré-series structure of the cumulative wall-crossing. The residue formula (10) controls the jump of $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ across a single pole-crossing. Summing such jumps along a continuous deformation of the contour pair from one that encloses no interior poles to one that encloses every pole in a single Γ -orbit $\Gamma \cdot \tau_p$ produces a cumulative residue of the structure

$$\sum_{\gamma \in \Gamma_{\tau_p} \backslash \Gamma} (c\tau_p + d)^{k-2} \text{Res}(f, \tau_p) \left[\sigma_\gamma^S (\gamma\tau_p)^{s-1} + \sigma_\gamma^T \tilde{k}_T(\gamma\tau_p, s) \right],$$

where the cocycle factor $(c\tau_p + d)^{k-2}$ comes from the transformation $\text{Res}(f, \gamma\tau_p) = (c\tau_p + d)^{k-2} \text{Res}(f, \tau_p)$ under modularity of f , Γ_{τ_p} is the stabiliser of τ_p in Γ , and $\sigma_\gamma^{S,T}$ encode the winding numbers of

$\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ across the orbit-point γ_{τ_p} . The summand is structurally a weight- k Poincaré-series term in τ_p , in the s -extended form built from τ_p^{s-1} and $\tilde{k}_T(\tau_p, s)$.

The structural parallel with the partial-fraction representation proved by Hardy and Ramanujan in their 1919 paper on modular functions with prescribed poles [12] is direct. [12, Thm. 2] shows that a modular function ϕ of weight n with a single simple pole at $\tau = \alpha$ admits the representation

$$f(q) = 2\pi i A \sum \frac{1}{(a + b\alpha)^{n+2}} \frac{\mathbf{q}^2}{q^2 - \mathbf{q}^2}, \quad \mathbf{q} = \exp\left(\frac{c+d\alpha}{a+b\alpha} \pi i\right),$$

summed over pairs (a, b) giving distinct Γ -images of α . Each summand is the residue at one orbit-point, transported by the modular cocycle factor $(a + b\alpha)^{-(n+2)}$ and weighted by an explicit kernel evaluated at $\mathbf{q}$. Our cumulative wall-crossing residue has the same orbit-of-residues structure, with $\tilde{k}_T(\gamma_{\tau_p}, s)$ playing the role of the $\mathbf{q}^2/(q^2 - \mathbf{q}^2)$ kernel. Whether the cumulative wall-crossing converges as a function of s to a canonical weight- k Poincaré series, and what its modular properties are, is a natural direction for follow-up work.

Towards systematic interior-pole L -function theory. Theorem 1.3 establishes existence and wall-crossing for $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ on arbitrary meromorphic f , but does not give a closed-form per-mode evaluation parallel to the incomplete-gamma sum (7). Developing the systematic per-mode theory — in particular adapting the polylog-projection machinery $\hat{P}_{\tau_p}$ (see [3]) to the finite-contour setting and matching the wall-crossing data (9) to the iterated-Eichler-integral framework of Brown–Hain on $\mathcal{M}_{1,1}$ and the universal elliptic curve [1] — is the natural sequel. The arithmetic of the wall-crossing residues $\mathcal{R}_{\tau_p}(f, s)$ for forms with poles at CM points (e.g. $\tau_p \in \{i, \rho\}$) is a particularly clean target, where the Laurent coefficients of f take values in number fields and $\mathcal{R}_{\tau_p}$ acquires explicit arithmetic structure.

A Algebraic derivation of the Hurwitz-zeta kernel

The body of §4.1 derives $\tilde{k}_T(\tau, s)$ by differentiating the two-segment functional in τ_0 and demanding the result vanish. Here we record the parallel derivation that proceeds *from* the classical Mellin transform of a holomorphic cusp form, splitting it at the contour endpoints $\{-1/\tau_0, \tau_0\}$ and demanding the resulting tail be representable as a T -segment integral. Both derivations produce the same shift-difference equation and the same kernel; this appendix makes the algebraic provenance of the Hurwitz-zeta combination manifest.

Step 1: Mellin split at the contour endpoints. Let $f \in S_k$ with k even. The classical completed L -function $L^*(f, s) = (2\pi)^{-s} \Gamma(s) L(f, s)$ has integral representation

$$L^*(f, s) = -i \int_0^{i\infty} f(\tau) (\tau/i)^{s-1} d\tau = i^{-s} \int_0^{i\infty} f(\tau) \tau^{s-1} d\tau,$$

the second equality using $(\tau/i)^{s-1} = i^{1-s} \tau^{s-1}$ on the imaginary axis. Pick $\tau_0 \in \mathbb{H}$ and split the imaginary-axis contour at the two endpoints $\{-1/\tau_0, \tau_0\}$ in (1). Reorganising the three resulting pieces using S -modularity $f(-1/\tau) = \tau^k f(\tau)$ on the lower segment $[0, -1/\tau_0]$ (via the substitution $\tau \mapsto -1/\tau$) and computing the q -series-against- τ^{s-1} integral on $[\tau_0, i\infty]$ directly yields, for every $\tau_0 \in \mathbb{H}$ and every $s \in \mathbb{C}$,

$$L^*(f, s) = i^{-s} \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^{s-1} d\tau + \sum_{n \geq 1} c_f(n) \left[\frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + \frac{\Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}} \right]. \quad (42)$$

The left side is τ_0 -independent (it is the classical L -value); both terms on the right side individually depend on τ_0 , but their sum does not. At $\tau_0 = i$ the argument $-2\pi i n \tau_0 = 2\pi n$ is real and (42) reduces to the standard BFK incomplete-gamma sum of [2].

The right side of (42) contains the S -segment integral $i^{-s} \int_{-1/\tau_0}^{\tau_0}$ explicitly; the remaining incomplete-gamma sum plays the role that we want to be a T -segment integral.

Step 2: ansatz for the T -segment kernel. Postulate that there exists a kernel $\tilde{k}_T(\tau, s)$, analytic in $\tau \in \mathbb{H}$ and meromorphic in $s \in \mathbb{C}$, such that

$$i^{-s} \int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau = \sum_{n \geq 1} c_f(n) \left[\frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + \frac{\Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}} \right] \quad (43)$$

for every $f \in S_k$ and every $\tau_0 \in \mathbb{H}$. If such a kernel exists, substituting (43) into (42) gives the two-segment functional $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = i^{-s} [\int_{\gamma_{\tau_0}^S} + \int_{\gamma_{\tau_0}^T}]$ as a τ_0 -independent expression equal to $L^*(f, s)$.

Step 3: τ_0 -derivative gives a shift-difference equation. Differentiate both sides of (43) in τ_0 . On the left, the only τ_0 -dependence is in the integration limits, so $d/d\tau_0$ acts as

$$\frac{d}{d\tau_0} \int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau = f(\tau_0) [\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s)],$$

using $f(\tau_0 - 1) = f(\tau_0)$ for the lower limit. On the right of (43), differentiating each incomplete gamma in its second argument (the explicit τ_0 -dependence) and assembling the Fourier series yields, after canceling a common factor of $f(\tau_0)$,

$$\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s) = -\tau_0^{s-1} + e^{i\pi(s-1)} \tau_0^{k-1-s} \quad \text{for all } \tau_0 \in \mathbb{H}. \quad (44)$$

The first term on the right is the contribution of the τ^{s-1} -Mellin endpoint at τ_0 ; the second term is the S -modularity image of the same endpoint at $-1/\tau_0$, carrying the branch factor $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)} \tau_0^{-(s-1)}$ together with the modularity Jacobian τ_0^k . Equation (44) is a shift-difference equation of order one in τ , with a sum of two power-of- τ terms on the right side.

Step 4: Hurwitz raising solves the shift-difference equation. The Hurwitz zeta function $\zeta(\sigma, a)$ satisfies the raising identity

$$\zeta(\sigma, a+1) - \zeta(\sigma, a) = -a^{-\sigma} \quad (\sigma \in \mathbb{C}, \operatorname{Re}(a) > 0) \quad (45)$$

(see [6, §25.11.3]). Applied at $(\sigma, a) = (1-s, \tau)$ and $(\sigma, a) = (s-(k-1), \tau)$ respectively, this gives

$$\zeta(1-s, \tau+1) - \zeta(1-s, \tau) = -\tau^{s-1}, \quad \zeta(s-(k-1), \tau+1) - \zeta(s-(k-1), \tau) = -\tau^{k-1-s}.$$

Combining with the relative phase $e^{i\pi(s-1)}$ from the second term of (44) gives the kernel

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1), \quad (46)$$

which by construction satisfies (44) and hence (43). The kernel is analytic in τ on $\mathbb{H}$ (both Hurwitz zetas have second argument $\tau+1$ with $\operatorname{Re}(\tau+1) > 0$) and meromorphic in s with simple poles only at $s=0$ and $s=k$, where the per-mode integrals collect removable cancellations (cf. Lemma 4.2). Equation (46) is the kernel (3) of the main text.

Where DR2022 stops. The kernel (46) has two Hurwitz-zeta terms. The construction of [8, Thm. 1.2] gives the single-term

$$\tilde{k}_T^{\text{DR}}(\tau, m) = \zeta^*(1 - m, \tau + 1)$$

on the segment $[i, i + 1]$, at integer m , which is the first term of (46). The structural origin of this gap is the absence of an S -segment in their construction: without the S -modularity-transformed contribution at the $-1/\tau_0$ endpoint in Step 3, the shift-difference equation (44) has only the single right-hand-side term $-\tau_0^{s-1}$, and its solution is the single-Hurwitz kernel. The second $e^{i\pi(s-1)}\tau_0^{k-1-s}$ term appears precisely because the two-segment functional includes the S -segment and its endpoint at $-1/\tau_0$ contributes under τ_0 -differentiation via S -modularity.

B Eichler–Shimura for physicists

This appendix is intended as an exposition aid for readers approaching the subject from theoretical physics. It is removable without affecting the logical content of the paper and is excluded from the journal/arXiv version. Primary references: Brown [1] §2–3 (clean modern exposition), Kohnen–Zagier 1984 [13] (classical), and Lewis–Zagier [14] (accessible motivation).

B.1 The polynomial coefficient system V_n and the slash action

Fix k even, set $n := k - 2$, and let V_n be the $\mathbb{Q}$ -vector space of polynomials in two formal variables X, Y , homogeneous of total degree n . $\Gamma = \text{SL}_2(\mathbb{Z})$ acts on V_n on the right by the *slash action*

$$(P|_\gamma)(X, Y) := P(aX + bY, cX + dY), \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \quad (47)$$

The two generators of Γ act as $P|_T(X, Y) = P(X + Y, Y)$ and $P|_S(X, Y) = P(-Y, X)$ on the basis monomials $X^{n-j}Y^j$.

B.2 Group cohomology $H^1(\Gamma; V_n)$

A 1-cocycle of Γ with coefficients in V_n is a map $C : \Gamma \rightarrow V_n$ satisfying

$$C_{gh} = C_g|_h + C_h, \quad g, h \in \Gamma. \quad (48)$$

This is exactly the property derived in Lemma 2.1. A 1-coboundary is one of the form $C_\gamma = P|_\gamma - P$ for some fixed $P \in V_n$. Coboundaries are automatically 1-cocycles. The *first group cohomology* is

$$H^1(\Gamma; V_n) := \{1\text{-cocycles}\} / \{1\text{-coboundaries}\}. \quad (49)$$

Because $\Gamma = \text{SL}_2(\mathbb{Z})$ is generated by S and T subject to the relations $S^2 = -I$, $(ST)^3 = -I$, a 1-cocycle is determined by the pair $(C_S, C_T) \in V_n \oplus V_n$ subject to the algebraic constraints

$$C_S|_S + C_S = 0, \quad C_T + C_{ST}|_{ST} + C_{ST}|_S = 0, \quad (50)$$

the latter coming from $(ST)^3 = -I$ together with $C_{(ST)^3} = C_{ST}|_{(ST)^2} + C_{ST}|_{ST} + C_{ST}$. Coboundaries $P|_\gamma - P$ contribute the pair $(P|_S - P, P|_T - P)$.

B.3 The Eichler integral cocycle

For $f \in S_k^!$ and basepoint $\tau_0 \in \mathbb{H}$, set

$$C_\gamma^f(\tau_0) := (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} f(\tau) (X - \tau Y)^n d\tau \in V_n \otimes \mathbb{C}. \quad (51)$$

This is the *Eichler integral cocycle* of f ; it is a 1-cocycle by Lemma 2.1. Its cohomology class $[C^f] \in H^1(\Gamma; V_n \otimes \mathbb{C})$ depends on f but not on τ_0 (for f without interior poles), because changing τ_0 adds a coboundary by (16).

B.4 The Eichler–Shimura theorem

The cohomology decomposes canonically as

$$H^1(\Gamma; V_n \otimes \mathbb{C}) \cong H_{\text{cusp}}^1(\Gamma; V_n \otimes \mathbb{C}) \oplus H_{\text{Eis}}^1(\Gamma; V_n \otimes \mathbb{C}), \quad (52)$$

with the cuspidal summand isomorphic to $S_k \oplus \overline{S_k}$ via the map $f \mapsto [C^f]_{\text{cusp}}$. In the $\dim S_k = 1$ substantive family $k \in \{12, 16, 18, 20, 22, 26\}$, the cuspidal summand is 2-dimensional, spanned by the two *cuspidal basis polynomials* $p_f^+, p_f^- \in V_n \otimes \mathbb{C}$ [1]. These are characterised by their behaviour under the S -action: p_f^+ satisfies $p_f^+|_S = p_f^+$ (even parity) and p_f^- satisfies $p_f^-|_S = -p_f^-$ (odd parity). Concretely, S acts on V_n by $(X, Y) \mapsto (-Y, X)$, so for $P = \sum_j p_j X^{n-j} Y^j$,

$$P|_S = \sum_j p_j (-Y)^{n-j} X^j = \sum_j (-1)^{n-j} p_j X^j Y^{n-j}; \quad (53)$$

that is, S interchanges $X \leftrightarrow Y$ and signs flip according to the monomial. The $\pm$ parity is then the action of this involution on $V_n \otimes \mathbb{C}$.

The Eisenstein summand H_{Eis}^1 is also non-trivial; for $\dim S_k = 1$, it is spanned by the Eisenstein coboundary class represented (up to scale) by the coboundary $\delta Y^n(\gamma) = Y^n|_\gamma - Y^n$. At $\gamma = S$, this gives $(-X)^n - Y^n = X^n - Y^n$ (using n even), the polynomial that contributes only to the boundary monomials X^n, Y^n of any representative cocycle.

B.5 Periods $\omega^\pm(f)$ and quasi-periods $\eta^\pm(f)$

For $f \in S_k$ (genuine holomorphic cusp form), the cuspidal projection of $[C^f]$ in the (p_f^+, p_f^-) basis defines two complex scalars,

$$[C^f]_{\text{cusp}} = \omega^+(f) p_f^+ + \omega^-(f) p_f^-, \quad (54)$$

called the *periods* of f . By the classical Manin/Eichler/Shimura identity, these encode the critical L -values $L(f, j) = \int_0^\infty t^{j-1} f(it) dt$ for integer j in the critical strip $\{1, \dots, k-1\}$.

For $f \in S_k^!$ with a cusp pole (e.g., $\widehat{\Delta} = q^{-1} + O(q^2) \in S_{12}^!$), the Eichler integral C^f is still a well-defined cocycle on a finite contour, and the cuspidal projection $[C^f]_{\text{cusp}} = \omega^+(f) p_f^+ + \omega^-(f) p_f^-$ defines a pair of scalars by the same formula. In this weakly holomorphic case they are denoted $\eta^\pm(f)$ and called *quasi-periods*; their relation to L -values requires the finite-contour regularisation studied in the main note (Theorem 1.1).

B.6 The Eisenstein-direction term in r_f versus the period polynomial of an Eisenstein series

A subtlety worth stating directly, in response to a common confusion. Two different objects are sometimes both called “period polynomials of Eisenstein origin”, and they have very different properties.

Object A: The Eisenstein direction in H^1 . In the decomposition (29) of r_f into cuspidal-basis polynomials $p_f^\pm$ plus an “Eisenstein-direction” term, the latter is by construction a multiple of the coboundary representative $Y^n|_S - Y^n = X^n - Y^n$ (or, equivalently, the polynomial $X^n - Y^n$ itself, which spans the Eisenstein direction in H^1). This contribution touches *only* the boundary monomials X^n and Y^n of r_f , leaving all interior monomials $X^{n-(\ell-1)}Y^{\ell-1}$, $\ell \in \{2, \dots, k-2\}$, free of Eisenstein contamination. This is what makes the interior-monomial extraction of $\omega^\pm(f)$ via (4) clean.

Object B: The period polynomial of an actual Eisenstein series. The period polynomial $r_{E_k}(X, Y)$ of the weight- k Eisenstein series $E_k(\tau)$ is a different object: it is the formal S -cocycle value of C^{E_k} modulo coboundaries, computed for $f = E_k$ itself. By Manin/Zagier (see [13] §2), r_{E_k} is supported on *all* monomials $X^j Y^{n-j}$ for $j \in \{0, 1, \dots, n\}$, with coefficients given by products of critical zeta values:

$$r_{E_k}(X, Y) \sim \sum_{j=0}^n \zeta^*(j+1) \zeta^*(k-1-j) \binom{n}{j} X^j Y^{n-j}, \quad (55)$$

where $\zeta^*(s) := \pi^{-s/2} \Gamma(s/2) \zeta(s)$ is the completed Riemann zeta. The factorisation $L(E_k, s) = \zeta(s) \zeta(s-k+1)$ sends *some* interior coefficients to zero — precisely those at j with $\zeta(j-k+2) = 0$, i.e., $j-k+2 \in \{-2, -4, -6, \dots\}$ (trivial zeros of ζ) — but *not* all interior coefficients. For $k = 12$, the vanishing happens at $j \in \{3, 5, 7, 9\}$ (and the parity-paired $j \in \{8, 6, 4, 2\}$ remain nonzero, modulo sign-pairing details). So $r_{E_{12}}$ is *not* supported only at the boundary monomials $j = 0, n$.

The distinction. Object A is a statement about *the structure of H^1* : the Eisenstein direction projects onto a 1-dimensional subspace whose representative coboundary touches only boundary monomials. Object B is a statement about *a specific cocycle representative* (the period polynomial of E_k), which generically has interior support. The two are not in conflict because what we actually use in the proof of Theorem 1.1 is Object A: the Eisenstein *direction’s* contribution to r_f at interior monomials vanishes, regardless of what the period polynomial of an actual Eisenstein series looks like.

References

- [1] F. Brown, *A class of non-holomorphic modular forms III*, arXiv:1710.07912 (2017).
- [2] K. Bringmann, K.-H. Fricke, and Z. Kent, *Special L -values and periods of weakly holomorphic modular forms*, Ramanujan J. **24** (2011), 119–139.
- [3] D. A. McGady, *L -functions for Meromorphic Modular Forms and Sum Rules in Conformal Field Theory*, arXiv:1806.09874 [hep-th] (2019); INSPIRE: <https://inspirehep.net/literature/1679799>.
- [4] Companion `mpmath` verification scripts (run at 50-digit precision unless noted), each probing a specific result of this note:
 - `verify_periods.py` — end-to-end verification of Theorem 1.1 for $f \in \{\Delta, \widehat{\Delta}\}$ at $k = 12$:
 - (a) reproduction of Brown’s $\omega^\pm(\Delta)$ and $\eta^\pm(\widehat{\Delta})$ [1] via the finite-contour cocycle pipeline;
 - (b) ℓ -independence across all nine interior monomials $X^{n-(\ell-1)}Y^{\ell-1}$, $\ell \in \{2, \dots, 10\}$;
 - (c) basepoint independence between $\tau_0 = 0.3 + 1.2i$ and $\tau_0 = -0.2 + 0.9i$;
 - (d) agreement of

the closed-form δ_T^{-1} of Lemma 3.2 with the strictly-lower-triangular forward substitution on every p_m ($m = 0, \dots, 9$); and (e) the polynomial identity of Lemma 3.5 — that $\widehat{k}_T(\ell, \tau) - (-1)^{\ell-1} \binom{n}{\ell-1} \widetilde{k}_T(\tau, \ell)$ is constant in $\mathbb{Q}$ — verified exactly via `sympy` at all nine interior ℓ .

- `dr_bLfunction_complex_s.py` — verification of the complex- s L -function of Theorem 1.2: the Hurwitz-extended functional (6) reproduces the canonical completed $L^*(\Delta, s)$ at half-integer, off-axis complex ($s = 2.7 + 0.4i$, $5 + i$, $5.5 + 2.5i$), and outside-strip ($s = 0.5, -0.3$) points to $\sim 10^{-53}$ — and the corresponding $L^*(\widehat{\Delta}, s)$ to $\sim 10^{-47}$ — together with τ_0 -independence at non-integer s .
- `verify_s6.py`, `verify_multi_s.py` — per-mode verification of Theorem 1.2 (the $\tau_0 = i$ collapse to the BFK incomplete-gamma sum) at $s \in \{1, 3, 6, 9, 11\}$, including the boundary cases $s = 1, 11$ outside the reach of the integer-monomial construction and the polar mode $n = -1$.

No script accompanies Theorem 1.3, which is established analytically in §5.

- [5] F. Brown, personal communication to the author, 28 June 2019. This correspondence describes the two-segment cocycle procedure (integrate along contours from $\gamma^{-1}\tau_0$ to τ_0 for the two generators $\gamma \in \{S, T\}$ of Γ , modify by a coboundary $P|(1 - T)$ to kill the T -component, then read $\omega^\pm$ and $\eta^\pm$ off the S -component), and is the structural starting point of the construction in the present note. An accompanying unpublished note on quasi-periods of cusp forms of levels 6 and 8 was attached to the correspondence.
- [6] NIST Digital Library of Mathematical Functions, F. W. J. Olver *et al.*, eds., <https://dlmf.nist.gov/>, Release 1.2.x. Chapters used: §8 (incomplete gamma) and §25.11 (Hurwitz zeta).
- [7] N. Diamantis and L. Rolén, *Period polynomials, derivatives of L -functions, and zeros of polynomials*, arXiv:1707.04814 (2017).
- [8] N. Diamantis and L. Rolén, *L -values of harmonic Maass forms*, arXiv:2201.10193 [math.NT] (2022); Trans. Amer. Math. Soc. **377** (2024), 3905–3926.
- [9] N. Diamantis, M. Lee, W. Raji, and L. Rolén, *L -series of harmonic Maass forms and a summation formula for harmonic lifts*, arXiv:2107.12366 [math.NT] (2021); Int. Math. Res. Not. IMRN **2023**, no. 18, 15729–15765.
- [10] Ö. Imamoglu, Y. Martin, and Á. Tóth, *An analogue of the Mellin transform for weakly holomorphic cusp forms and a converse theorem*, preprint (2024).
- [11] J. H. Bruinier, J. Funke, and Ö. Imamoglu, *Regularized theta liftings and periods of modular functions*, arXiv:1112.3444 [math.NT] (2011); J. Reine Angew. Math. **703** (2015), 43–93.
- [12] G. H. Hardy and S. Ramanujan, *On the coefficients in the expansions of certain modular functions*, Proc. Roy. Soc. Lond. A **95** (1919), 144–155.
- [13] W. Kohnen and D. Zagier, *Modular forms with rational periods*, in *Modular Forms* (R. A. Rankin, ed.), Ellis Horwood, 1984, pp. 197–249.
- [14] J. Lewis and D. Zagier, *Period functions for Maass wave forms. I*, Ann. of Math. **153** (2001), 191–258.

- [15] Y.-t. Huang and D. McGady, *Consistency Conditions for Gauge Theory S Matrices from Requirements of Generalized Unitarity*, Phys. Rev. Lett. **112** (2014) 241601, arXiv:1307.4065 [hep-th].
- [16] W.-M. Chen, Y.-t. Huang, and D. A. McGady, *Anomalies without an action*, arXiv:1402.7062 [hep-th] (2014).